

ISC Class XII Mathematics

Chapter 2

Inverse Trigonometric Functions

51 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

1. Principal values & direct evaluation of inverse trig functions
2. Reducing $\sin^{-1}(\sin x)$ -type expressions outside the principal branch
3. Domain, range & maximum-value questions
4. Right-triangle ratio method (given one inverse trig value, find another)
5. Composite evaluation: trig of an inverse trig value
6. Half-angle substitution (Example 11 method)
7. Complementary identity family ($\sin^{-1} x + \cos^{-1} x = \pi/2$, etc.)
8. Simplification by trigonometric substitution ($x = \tan \theta, \sin \theta$, etc.)
9. Solving simple inverse-trig equations for x
10. Addition/subtraction formula: $\tan^{-1} a + \tan^{-1} b$ and its variants
11. Proofs using the addition formula
12. Boundary-case and extreme-value type (e.g. $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$)
13. Double-angle recognition: $2t/(1+t^2)$ and $(1-t^2)/(1+t^2)$ patterns
14. Quadratic equations in x from the addition formula (incl. ISC 2023)
15. Nested composite equations ($\cos^{-1}(\sin(\cos^{-1} x))$ type)

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Type	Method / Pattern	★	Source(s)
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Formula Sheet

Principal Value Ranges

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \cos^{-1} x \in [0, \pi], \tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$x \in [-1, 1]$ for sin/cos; $x \in \mathbb{R}$ for tan

Principal Value Ranges

$$\cot^{-1} x \in (0, \pi), \sec^{-1} x \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}, \operatorname{cosec}^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$$

$x \in \mathbb{R}$ for cot; $|x| \geq 1$ for sec/cosec

Cancellation Identities

$$\sin(\sin^{-1} x) = x, \cos(\cos^{-1} x) = x, \tan(\tan^{-1} x) = x$$

Valid on the stated domain of each function

Reciprocal Identities

$$\sin^{-1} x = \operatorname{cosec}^{-1}\left(\frac{1}{x}\right), \cos^{-1} x = \sec^{-1}\left(\frac{1}{x}\right), \tan^{-1} x = \cot^{-1}\left(\frac{1}{x}\right) \quad (x > 0)$$

Sign flips by $-\pi$ or π when $x < 0$ for the cot version

Complementary Identities

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \quad \sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$$

All valid on the common domain of each pair

Odd Function Identities

$$\sin^{-1}(-x) = -\sin^{-1} x, \tan^{-1}(-x) = -\tan^{-1} x, \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

x in the respective domain

Supplementary Identities

$$\cos^{-1}(-x) = \pi - \cos^{-1} x, \cot^{-1}(-x) = \pi - \cot^{-1} x, \sec^{-1}(-x) = \pi - \sec^{-1} x$$

x in the respective domain

Addition Formula

$$\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a + b}{1 - ab} \right)$$

Valid directly when $ab < 1$; add/subtract π if $ab > 1$ (sign of a, b decides)

Subtraction Formula

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a - b}{1 + ab} \right)$$

Valid for all real a, b

Double Angle (tan form)

$$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1 - x^2} \right)$$

Valid for $|x| < 1$

Double Angle (sin form)

$$2 \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1 + x^2} \right)$$

Valid for $|x| \leq 1$

Double Angle (cos form)

$$2 \tan^{-1} x = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right)$$

Valid for $x \geq 0$

Substitution: x

$$\tan^{-1} \left(\frac{x}{\sqrt{1 - x^2}} \right) = \sin^{-1} x, \quad \sin^{-1} \left(\frac{x}{\sqrt{1 + x^2}} \right) = \tan^{-1} x$$

Used to simplify algebraic expressions under an inverse trig function

Substitution: x

$$\operatorname{cosec}^{-1} \left(\frac{\sqrt{1 + x^2}}{x} \right) = \tan^{-1} x, \quad \cot^{-1} \left(\frac{x}{\sqrt{1 - x^2}} \right) = \sin^{-1} x$$

Companion forms of the same substitution trick

Half-Angle (Example 11 method)

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}, \quad \sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$$

Used after setting $\frac{1}{2} \sin^{-1}(\cdot)$ or $\frac{1}{2} \cos^{-1}(\cdot) = x$

Periodicity Reduction

$$\tan^{-1}(\tan \theta) = \theta - n\pi, \quad \sin^{-1}(\sin \theta) = (-1)^n(\theta - n\pi), \quad \cos^{-1}(\cos \theta) = \pm\theta + 2n\pi$$

Choose integer n so the reduced angle lands inside the principal branch

I. Principal Values & Direct Evaluation

Type 1

Evaluate $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$, $\cot^{-1}$, $\sec^{-1}$ or $\operatorname{cosec}^{-1}$ of a standard trig value, reporting the answer inside the correct principal-value range. The version that actually catches students out is a *negative* standard value, where the odd/supplementary rule for that specific function must be applied correctly – not just recalled from a positive-value table.

Source: Exercise 2.1, Q4(i)

Given: $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

Solution:

$\cot^{-1}$ has principal range $(0, \pi)$, unlike $\tan^{-1}$'s symmetric range – so a negative input does *not* simply give a negative angle here; it lands in the second quadrant instead.

Since $\cot \frac{\pi}{3} = \frac{1}{\sqrt{3}}$, the supplementary rule for cotangent gives $\cot\left(\pi - \frac{\pi}{3}\right) = -\cot \frac{\pi}{3} = -\frac{1}{\sqrt{3}}$, so the required angle is $\pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Check: $\frac{2\pi}{3} \in (0, \pi)$, so this is a valid principal value.

Final Answer

$$\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \frac{2\pi}{3}$$

Common Mistake: Applying the sin/tan odd-function rule ($f^{-1}(-x) = -f^{-1}(x)$) to $\cos^{-1}$, $\cot^{-1}$, or $\sec^{-1}$ – these three instead use the *supplementary* rule $f^{-1}(-x) = \pi - f^{-1}(x)$, since their principal ranges don't straddle zero symmetrically.

Quick Recall: Before evaluating a negative standard value, check which range family the function belongs to: sin/tan/cosec are odd ($-\theta$), while cos/cot/sec are supplementary ($\pi - \theta$) – picking the wrong rule is the single most common slip in this whole topic.

Type 2

Evaluate $f^{-1}(f(x))$ (e.g. $\sin^{-1}(\sin x)$, $\tan^{-1}(\tan x)$) where the inner angle x lies *outside* the principal branch of f^{-1} , so the naive answer " $= x$ " is wrong. The trick is to write x as (principal-branch angle) $\pm n\pi$ using periodicity, so that the trig value is unchanged but the angle used is inside the principal branch.

★ **Board Favorite** – This exact pattern (reducing $\sin^{-1}(\sin x)$ / $\tan^{-1}(\tan x)$ / $\sec^{-1}(\sec x)$ for x outside the principal branch) is one of the most consistently tested ISC patterns on this chapter – it appears almost every year in some form because it directly tests whether students understand principal value as a restriction, not just a formula.

Formula Used: $\cos^{-1}(\cos \theta) = \theta$ only if $\theta \in [0, \pi]$; otherwise write $\theta = 2n\pi \pm \theta'$ with $\theta' \in [0, \pi]$

Source: Exercise 2.1, Q5(i)

Given: $\cos^{-1}\left(\cos \frac{4\pi}{3}\right)$

Solution:

The principal branch of $\cos^{-1}$ is $[0, \pi]$, and $\frac{4\pi}{3}$ does not lie in this interval (it's bigger than π), so we cannot just cancel and write the answer as $\frac{4\pi}{3}$ – that would be outside the allowed range.

We need to find an angle in $[0, \pi]$ that has the same cosine value as $\frac{4\pi}{3}$. Using the identity $\cos(2\pi - \theta) = \cos \theta$ is one route, but here it's cleaner to use $\cos\left(2\pi - \frac{4\pi}{3}\right) = \cos \frac{4\pi}{3}$, giving $\cos \frac{2\pi}{3} = \cos \frac{4\pi}{3}$.

Since $\frac{2\pi}{3}$ lies in $[0, \pi]$, this is now a valid principal-branch angle, so we can cancel $\cos^{-1}(\cos(\cdot))$ safely.

Final Answer

$$\cos^{-1}\left(\cos \frac{4\pi}{3}\right) = \frac{2\pi}{3}$$

Common Mistake: Writing $\cos^{-1}(\cos \theta) = \theta$ automatically without first checking θ lies in $[0, \pi]$ – the single most common error across this whole type, for all six inverse functions.

Alternate Board-Style Example

Source: Exercise 2.1, Q6(ii)

Given: $\tan^{-1}\left(\tan \frac{9\pi}{8}\right)$

Solution:

The principal branch of $\tan^{-1}$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and $\frac{9\pi}{8}$ is well outside this range, so direct cancellation is invalid.

Since $\tan$ has period π , we subtract π : $\tan\left(\frac{9\pi}{8} - \pi\right) = \tan \frac{\pi}{8} = \tan \frac{9\pi}{8}$, and this preserves the tangent value while shifting the angle by a full period.

Now $\frac{\pi}{8}$ lies inside $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so the cancellation is valid at this angle.

Final Answer

$$\tan^{-1}\left(\tan \frac{9\pi}{8}\right) = \frac{\pi}{8}$$

Quick Recall: Before cancelling $f^{-1}(f(\theta)) = \theta$, always check θ is inside the principal branch – if not, add/subtract the right multiple of the period (2π for $\sin/\cos$, π for $\tan/\cot$) until it lands inside, keeping the trig value unchanged.

Type 3

A "show that" question asking you to prove $\tan^{-1}(\tan \theta) \neq \theta$ for a specific θ outside the principal branch, and then state what the actual value is – this is Type 2's method but framed as a proof rather than a plain evaluation.

Formula Used: $\tan(\theta - \pi) = \tan \theta$ (period π)

Source: Exercise 2.1, Q7(i)

Given: Show that $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$, and find its actual value.

Solution:

First we confirm $\frac{5\pi}{6}$ is outside the principal branch $(-\frac{\pi}{2}, \frac{\pi}{2})$ of $\tan^{-1}$ (since $\frac{5\pi}{6} > \frac{\pi}{2}$), which is exactly why the naive cancellation fails – this is the "show that" part of the question. Using periodicity, $\tan\left(\frac{5\pi}{6} - \pi\right) = \tan\left(-\frac{\pi}{6}\right) = \tan \frac{5\pi}{6}$, so the tangent value is unchanged when we shift by π .

Now $-\frac{\pi}{6}$ lies inside $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6}$.

Final Answer

$$\tan^{-1}\left(\tan \frac{5\pi}{6}\right) = -\frac{\pi}{6}, \text{ which is clearly } \neq \frac{5\pi}{6}$$

Common Mistake: Stopping after showing the inequality without actually computing the correct value – the question asks for both.

Quick Recall: "Show that $\neq$ " questions are Type 2 in disguise: prove the angle is outside the branch, then do the same periodicity shift to find the real answer.

Type 4

Find the domain of a composite expression involving one or more inverse trig functions – you must intersect the domain restrictions of every inverse trig function used, plus any restriction from functions like square roots wrapped around them.

★ **Board Favorite** – Domain questions on inverse trig functions are a standard high-yield pattern for this chapter, especially when combined with a second function like $\sin x$ or a square root, since they test whether students know the domains of all six inverse functions, not just $\sin^{-1}$ and $\cos^{-1}$.

Formula Used: $\text{Domain}(\sin^{-1} x) = \text{Domain}(\cos^{-1} x) = [-1, 1]; \text{Domain}(\sec^{-1} x) = \text{Domain}(\text{cosec}^{-1} x) = (-\infty, -1] \cup [1, \infty)$

Source: Exercise 2.1, Q8(ii)

Given: $\sec^{-1}(2x - 3)$

Solution:

The inner argument of $\sec^{-1}$ must satisfy $|2x - 3| \geq 1$ since $\sec^{-1}$ is only defined for inputs of magnitude at least 1 – this comes directly from the domain of $\sec^{-1}$.

Splitting the absolute value: either $2x - 3 \geq 1$, giving $x \geq 2$, or $2x - 3 \leq -1$, giving $x \leq 1$ – we solve both branches of the inequality separately.

Combining the two branches (either one is acceptable, since we need $|2x - 3| \geq 1$, not both conditions simultaneously) gives the full domain.

Final Answer

Domain = $(-\infty, 1] \cup [2, \infty)$

Common Mistake: Forgetting that $\sec^{-1}$ and cosec^{-1} exclude the open interval $(-1, 1)$, not just the point 0 – students often write $|2x - 3| \neq 0$ instead of the correct $|2x - 3| \geq 1$.

Alternate Board-Style Example

Source: Exercise 2.1, Q8(iii)

Given: $\cos^{-1} x + \sin x$

Solution:

$\cos^{-1} x$ requires $x \in [-1, 1]$, while $\sin x$ is defined for every real number, so it places no extra restriction on the domain.

Since we need both parts of the sum to be defined simultaneously, we intersect the two domains: $[-1, 1] \cap \mathbb{R} = [-1, 1]$.

Final Answer

Domain = $[-1, 1]$

Quick Recall: List the domain restriction of every inverse trig piece separately, then intersect them all – the domain of the whole expression is only as wide as its most restrictive piece.

Type 5

A short conceptual question asking you to state the range of a *non-principal* branch of $\sin^{-1}$ or $\cos^{-1}$ – this tests understanding that inverse trig functions are multi-branched, and the principal branch is just one conventional choice among infinitely many.

Source: Exercise 2.1, Q9(i)

Given: Write the range of one branch of $\sin^{-1} x$, other than the principal branch.

Solution:

The principal branch of $\sin^{-1} x$ has range $[-\frac{\pi}{2}, \frac{\pi}{2}]$; every other branch is obtained by shifting this interval by a multiple of π (with alternating orientation, since $\sin$ repeats with a sign-flip pattern every π).

The very next branch, shifting by one period, has range $[\frac{\pi}{2}, \frac{3\pi}{2}]$ – any consecutive interval of length π other than $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is an acceptable answer here.

Final Answer

One valid non-principal branch: range = $[\frac{\pi}{2}, \frac{3\pi}{2}]$

Common Mistake: Giving an interval of the wrong length (not π) or an interval that overlaps two branches instead of describing exactly one clean branch.

Quick Recall: Every inverse trig function has infinitely many valid branches, each an interval of the same length as the principal one, tiled end-to-end along the real line.

Type 6

Find the maximum (or minimum) value of an expression like $(\cos^{-1} x)^2$ by using the known range of the inner inverse trig function.

Source: Exercise 2.1, Q10

Given: Find the maximum value of $(\cos^{-1} x)^2$

Solution:

We know $\cos^{-1} x \in [0, \pi]$ for every valid $x \in [-1, 1]$, so we are squaring a quantity that ranges over this interval.

Since $\cos^{-1} x \geq 0$ always, squaring is an increasing operation on this range, so the square is largest exactly when $\cos^{-1} x$ itself is largest, i.e. at $\cos^{-1} x = \pi$ (which happens at $x = -1$).

Final Answer

Maximum value $= \pi^2$ (attained at $x = -1$)

Common Mistake: Assuming the maximum of a squared expression occurs where the inner value has the largest magnitude in either direction – here that reasoning happens to work because $\cos^{-1} x \geq 0$ always, but students often try to also check $x = 1$ (giving 0) as a candidate for the maximum instead of the minimum.

Quick Recall: Find the range of the inner inverse trig function first, then reason about how the outer operation (squaring, etc.) behaves on that specific range.

Type 7

Given that $\cot^{-1} \left(\frac{p}{q} \right) = x$ (or the same for $\tan^{-1}$), find $\cos x$ and $\sin x$ directly, by imagining a right triangle with the given ratio as two of its sides and using Pythagoras for the third.

★ **Board Favorite** – This right-triangle-ratio technique is a standard high-yield pattern for this chapter – it recurs constantly as a sub-step inside larger problems (addition formula proofs, half-angle questions), so mastering it directly pays off across the whole chapter, not just this question type.

Formula Used: If $\tan x = \frac{p}{q}$ (or $\cot x = \frac{p}{q}$), build a right triangle with legs p, q and hypotenuse $\sqrt{p^2 + q^2}$

Source: Exercise 2.1, Q11

Given: $\cot^{-1} \left(\frac{3}{4} \right) = x$, find $\cos x$

Solution:

Since $\cot^{-1} \left(\frac{3}{4} \right) = x$, we have $\cot x = \frac{3}{4}$, which means in a right triangle, the side adjacent to angle x is 3 and the side opposite is 4 – this comes directly from what cotangent means (adjacent/opposite).

By Pythagoras, the hypotenuse is $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$ – we need this third side to build $\cos x$, which needs the hypotenuse.

Since $\cot^{-1}$ has range $(0, \pi)$, x is an angle in the first or second quadrant, and here $\cot x > 0$ forces x into the first quadrant, so $\cos x$ is positive.

Therefore $\cos x = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{3}{5}$.

Final Answer

$$\cos x = \frac{3}{5} \text{ (and, by the same triangle, } \sin x = \frac{4}{5}\text{)}$$

Common Mistake: Forgetting to check which quadrant x actually lies in based on the range of the inverse function used, and just assuming all ratios are positive – this causes sign errors when the given value is negative.

Alternate Board-Style Example

Source: Exercise 2.1, Q12(ii)

Given: $\cot^{-1}\left(-\frac{1}{7}\right) = x$, find $\sin x$ and $\cos x$

Solution:

Here $\cot x = -\frac{1}{7}$ is negative, and since $\cot^{-1}$ has range $(0, \pi)$, a negative cotangent places x in the second quadrant (where sine is positive but cosine is negative).

Building the triangle with the magnitudes 1 and 7, the hypotenuse is $\sqrt{1^2 + 7^2} = \sqrt{50} = 5\sqrt{2}$.

Applying the correct quadrant signs: $\sin x = \frac{7}{5\sqrt{2}}$ (positive) and $\cos x = -\frac{1}{5\sqrt{2}}$ (negative, since x is in the second quadrant).

Final Answer

$$\sin x = \frac{7}{5\sqrt{2}}, \quad \cos x = -\frac{1}{5\sqrt{2}}$$

Quick Recall: Turn the given ratio into a right triangle, find the missing side with Pythagoras, then fix the signs using which quadrant the range of that inverse function actually restricts you to.

Type 8

An equation of the form $\tan^{-1} x = \sin^{-1}(a)$ (two different inverse trig functions set equal) – solved by converting both sides into the same trig function of the same underlying angle, usually through a right triangle.

Source: Exercise 2.1, Q12(iii)

Given: $\tan^{-1} x = \sin^{-1}\left(\frac{1}{2}\right)$, find x

Solution:

Let $\sin^{-1}\left(\frac{1}{2}\right) = \theta$, so $\sin \theta = \frac{1}{2}$, and since this is a standard value, $\theta = \frac{\pi}{6}$ – converting the

right side into a concrete angle first makes the rest of the problem straightforward. The equation now reads $\tan^{-1} x = \frac{\pi}{6}$, so taking $\tan$ of both sides gives $x = \tan \frac{\pi}{6}$.

Final Answer

$$x = \frac{1}{\sqrt{3}}$$

Common Mistake: Trying to directly equate the "inputs" ($x = \frac{1}{2}$) instead of recognising that $\tan^{-1}$ and $\sin^{-1}$ of the same numeric input generally give different angles – the two sides must first be reduced to actual angles before comparing.

Quick Recall: When two different inverse trig functions are set equal, convert the side with a standard/nice value into an actual angle first, then apply the other function to that angle.

Type 9

Evaluate a composite expression where the inner value is *not* a standard angle, so it must be handled via a right-triangle ratio rather than a memorised value – and where the outer expression involves more than one inverse trig term, forcing you to track two different triangles at once.

★ **Board Favorite** – This exact question appeared on the ISC 2020 board paper – it's a genuine board favourite precisely because it forces two different inverse trig conversions inside a single expression, rather than testing one in isolation.

Formula Used: $\cos 2\theta = 2\cos^2 \theta - 1$

Source: ISC 2020 Board Paper

Given: Evaluate $\cos(2\cos^{-1} x + \sin^{-1} x)$ at $x = \frac{1}{5}$

Solution:

Rewrite the angle as $2\cos^{-1} x + \sin^{-1} x = \cos^{-1} x + (\cos^{-1} x + \sin^{-1} x) = \cos^{-1} x + \frac{\pi}{2}$, using the complementary identity to collapse one $\cos^{-1} x$ and one $\sin^{-1} x$ pair.

So $\cos(2\cos^{-1} x + \sin^{-1} x) = \cos\left(\cos^{-1} x + \frac{\pi}{2}\right) = -\sin(\cos^{-1} x)$, using $\cos\left(\theta + \frac{\pi}{2}\right) = -\sin \theta$.

Build the triangle for $\cos^{-1} x$: adjacent = x , hypotenuse = 1, so opposite = $\sqrt{1-x^2}$, giving $\sin(\cos^{-1} x) = \sqrt{1-x^2}$.

At $x = \frac{1}{5}$: $\sqrt{1 - \frac{1}{25}} = \sqrt{\frac{24}{25}} = \frac{2\sqrt{6}}{5}$.

Final Answer

$$\cos\left(2\cos^{-1}\frac{1}{5} + \sin^{-1}\frac{1}{5}\right) = -\frac{2\sqrt{6}}{5}$$

Common Mistake: Expanding $\cos(2\cos^{-1}x + \sin^{-1}x)$ directly via the full cosine addition formula (treating it as three separate angles) – this works but is far slower under exam time pressure than first collapsing the $\cos^{-1}x + \sin^{-1}x$ pair to $\pi/2$.

Quick Recall: Whenever an expression contains both $\sin^{-1}x$ and $\cos^{-1}x$ of the *same* x , look for a way to pair them into $\pi/2$ first – it almost always shortens the whole problem before any triangle is even needed.

Type 10

Prove that a sum (or combination) of two or more inverse trig values, each a standard angle, equals a stated numeric angle – essentially a "show that" version of Type 1 applied twice and added.

Source: Exercise 2.1, Q14(i)

Given: Show that $\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{2\pi}{3}$

Solution:

We evaluate each term separately using standard angles: $\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$ (since $\sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$ and $-\frac{\pi}{6}$ lies in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

Next, $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}$, using the supplementary identity $\cos^{-1}(-x) = \pi - \cos^{-1}x$ applied to $\cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{6}$, giving $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

Adding the two results: $-\frac{\pi}{6} + \frac{5\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3}$, which matches the value we were asked to prove.

Final Answer

$$\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{2\pi}{3} \text{ (verified)}$$

Common Mistake: Forgetting the negative sign flips for $\sin^{-1}(-x)$ (odd function, so answer is negative) versus $\cos^{-1}(-x)$ (supplementary, so answer is π minus the positive-value answer) – these two rules are opposite in form and easy to mix up.

Quick Recall: Evaluate each inverse trig term to a standard angle using the odd/supplementary rules as needed, then simply add or subtract the resulting angles.

Type 11

"Using principal values, find the value of" a linear combination (sum/difference, possibly with integer coefficients) of several inverse trig terms, each evaluated at a standard value.

Source: Exercise 2.1, Q15(iii)

Given: $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right)$

Solution:

Evaluate each term at its standard angle: $\tan^{-1}(1) = \frac{\pi}{4}$, since $\tan \frac{\pi}{4} = 1$ and $\frac{\pi}{4}$ is inside the principal branch.

$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$, using the supplementary identity on the standard value $\cos^{-1} \frac{1}{2} = \frac{\pi}{3}$, giving $\pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Adding: $\frac{\pi}{4} + \frac{2\pi}{3} = \frac{3\pi}{12} + \frac{8\pi}{12} = \frac{11\pi}{12}$.

Final Answer

$$\frac{11\pi}{12}$$

Common Mistake: Arithmetic slip when adding fractions with different denominators under π – always convert to a common denominator carefully before adding.

Quick Recall: This is just Type 1 repeated for every term in the expression, followed by ordinary fraction arithmetic on the resulting angles.

Type 12

A nested expression like $\tan^{-1}\left(2 \cos\left(2 \sin^{-1} \frac{1}{2}\right)\right)$ – worked strictly from the innermost bracket outward, one layer at a time.

★ **Board Favorite** – Nested inverse-trig evaluations are flagged as "Competency Focused" in this book's own exercise labelling and are a standard high-yield pattern, since they test careful, ordered evaluation rather than a single formula.

Formula Used: Work strictly innermost-to-outermost; evaluate one layer completely before moving

Source: Exercise 2.1, Q16(i)

Given: $\tan^{-1}\left(2 \cos\left(2 \sin^{-1} \frac{1}{2}\right)\right)$

Solution:

Innermost layer first: $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$, a standard value.

Next layer: $2 \times \frac{\pi}{6} = \frac{\pi}{3}$, so we now need $\cos \frac{\pi}{3}$.

$\cos \frac{\pi}{3} = \frac{1}{2}$, another standard value, so the expression inside the outer $2(\cdot)$ becomes $\frac{1}{2}$.

Next layer: $2 \times \frac{1}{2} = 1$, so finally we need $\tan^{-1}(1)$.

$\tan^{-1}(1) = \frac{\pi}{4}$, completing the outermost layer.

Final Answer

$$\frac{\pi}{4}$$

Common Mistake: Trying to simplify multiple layers at once instead of working strictly one layer at a time – this is where sign or standard-value errors creep in on nested problems.

Alternate Board-Style Example

Source: Exercise 2.1, Q16(iv)

Given: $\cot(\sin^{-1}(\cos(\tan^{-1} 1)))$

Solution:

Innermost: $\tan^{-1} 1 = \frac{\pi}{4}$.

Next: $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$.

Next: $\sin^{-1} \frac{1}{\sqrt{2}} = \frac{\pi}{4}$.

Finally: $\cot \frac{\pi}{4} = 1$.

Final Answer

$$1$$

Quick Recall: Peel the expression like an onion – solve the innermost bracket completely to a number before touching the next layer out, never combine layers.

Type 13

A sum of several $\tan^{-1}(\tan \cdot)$ / $\cos^{-1}(\cos \cdot)$ / $\sec^{-1}(\sec \cdot)$ terms, each needing its own periodicity reduction (Type 2), combined into a single final expression.

Source: Exercise 2.1, Q18(i)

Given: $\sin^{-1} \left(\sin \frac{3\pi}{4} \right) + \cos^{-1}(\cos \pi) + \tan^{-1}(1)$

Solution:

First term: $\frac{3\pi}{4}$ is outside $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so we reduce using $\sin(\pi - \theta) = \sin \theta$: $\sin(\pi - \frac{3\pi}{4}) = \sin \frac{\pi}{4} = \sin \frac{3\pi}{4}$, and $\frac{\pi}{4}$ is inside the branch, so $\sin^{-1}(\sin \frac{3\pi}{4}) = \frac{\pi}{4}$.

Second term: π already lies in $[0, \pi]$, the principal branch of $\cos^{-1}$, so $\cos^{-1}(\cos \pi) = \pi$.

directly, no reduction needed.

Third term: $\tan^{-1}(1) = \frac{\pi}{4}$, a direct standard value.

Adding all three: $\frac{\pi}{4} + \pi + \frac{\pi}{4} = \frac{3\pi}{2}$.

Final Answer

$$\frac{3\pi}{2}$$

Common Mistake: Applying a reduction step even to terms that are already inside the principal branch (like the $\cos^{-1}(\cos \pi)$ term here) – always check first whether reduction is even needed for each individual term.

Quick Recall: Handle each term completely on its own using Type 2's method, then simply add the resulting angles at the end.

Type 14

Evaluate $\text{trig}(\text{inverse-trig}(\text{fraction}))$ where the fraction corresponds to a Pythagorean triple (3-4-5, 5-12-13, etc.), letting you read off the answer directly without drawing anything, once the triple is recognised.

Formula Used: If $\sin^{-1} \frac{p}{r} = \theta$ with (p, q, r) a Pythagorean triple, then $\cos \theta = \frac{q}{r}$ directly

Source: Exercise 2.1, Q19(ii)

Given: $\text{cosec} \left(\cos^{-1} \left(-\frac{12}{13} \right) \right)$

Solution:

We recognise 5, 12, 13 as a Pythagorean triple, so the pair 12, 13 instantly tells us the third side is 5, without needing to compute $\sqrt{13^2 - 12^2}$ from scratch.

Since the input to $\cos^{-1}$ is negative and $\cos^{-1}$ has range $[0, \pi]$, the angle lies in the second quadrant, where sine (and hence cosecant) is still positive.

Using the triple, $\sin \theta = \frac{5}{13}$ (positive, as reasoned above), so $\text{cosec} \theta = \frac{13}{5}$.

Final Answer

$$\frac{13}{5}$$

Common Mistake: Getting the sign wrong for the second quadrant – students sometimes assume a negative input to $\cos^{-1}$ must make everything negative, but sine (and cosecant) stay positive throughout $[0, \pi]$.

Quick Recall: Spot the Pythagorean triple in the numbers, use it to get the missing side instantly, then apply the correct sign based on which quadrant that particular inverse function's range restricts you to.

Type 15

The signature "half-angle" type: show that $\tan\left(\frac{1}{2}\sin^{-1}x\right)$ or $\tan\left(\frac{1}{2}\cos^{-1}x\right)$ equals a given surd expression. Solved by setting the half-angle equal to a variable, doubling it back using the double-angle cosine formula in terms of $\tan x$, and solving the resulting quadratic in $\tan x$ – exactly the method shown in Example 11 of this chapter.

★ **Board Favorite** – This half-angle technique (Example 11's method) is a standard high-yield pattern for this chapter – it is explicitly worked as a "Competency Focused Question" in the textbook itself, signalling it is considered important enough to test directly.

Formula Used: $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$

Formula Used: Restrict the sign of $\tan x$ using the known range of x forced by the half-angle substitution.

Source: Exercise 2.1, Q20

Given: Show that $\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \frac{4 - \sqrt{7}}{3}$

Solution:

Let $\frac{1}{2}\sin^{-1}\frac{3}{4} = x$, so $\sin^{-1}\frac{3}{4} = 2x$, and since $\sin^{-1}$ has range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we get $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$ after halving – this range restriction on x is essential later for choosing the correct root. Taking sine of both sides: $\sin 2x = \frac{3}{4}$.

We convert this to a statement about $\tan x$ using $\sin 2x = \frac{2 \tan x}{1 + \tan^2 x} = \frac{3}{4}$, since our target expression is in terms of $\tan x$, not $\sin x$ or $\cos x$ directly.

Cross-multiplying: $8 \tan x = 3(1 + \tan^2 x)$, giving the quadratic $3 \tan^2 x - 8 \tan x + 3 = 0$.

Using the quadratic formula: $\tan x = \frac{8 \pm \sqrt{64 - 36}}{6} = \frac{8 \pm \sqrt{28}}{6} = \frac{4 \pm \sqrt{7}}{3}$ – this gives two candidate roots, and we must pick the one consistent with the range of x .

Since $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$, we need $-1 \leq \tan x \leq 1$; checking, $\frac{4 + \sqrt{7}}{3} \approx 2.22$ is too large, so it is rejected, while $\frac{4 - \sqrt{7}}{3} \approx 0.45$ fits inside $[-1, 1]$ and is the valid root.

Final Answer

$$\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \frac{4 - \sqrt{7}}{3}$$

Common Mistake: Accepting both roots of the quadratic without checking which one is actually consistent with the restricted range of x forced by the half-angle substitution – this range check is not optional, it's the step that decides the final answer.

Alternate Board-Style Example

Source: Textbook Example 11(ii)

Given: Evaluate $\tan \left(\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3} \right)$

Solution:

Let $\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3} = x$, so $\cos^{-1} \frac{\sqrt{5}}{3} = 2x$, and since $\cos^{-1}$ has range $[0, \pi]$, halving gives $0 \leq x \leq \frac{\pi}{2}$, which forces $\tan x \geq 0$.

Taking cosine of both sides: $\cos 2x = \frac{\sqrt{5}}{3}$, and converting via $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$ leads,

after cross-multiplying and simplifying, to $\tan^2 x = \frac{3 - \sqrt{5}}{3 + \sqrt{5}}$.

Rationalising the denominator: $\tan^2 x = \frac{(3 - \sqrt{5})^2}{(3 + \sqrt{5})(3 - \sqrt{5})} = \frac{9 - 6\sqrt{5} + 5}{4} = \frac{14 - 6\sqrt{5}}{4} =$

$$\frac{7 - 3\sqrt{5}}{2}.$$

Since $\tan x \geq 0$ from our range check, we take the positive square root: after simplification this equals $\frac{3 - \sqrt{5}}{2}$ (the positive root consistent with $0 \leq x \leq \frac{\pi}{2}$).

Final Answer

$$\tan \left(\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3} \right) = \frac{3 - \sqrt{5}}{2}$$

Quick Recall: Set the half-angle as x , double it back with the tan-form of the double-angle formula, solve the resulting quadratic in $\tan x$, then reject whichever root doesn't fit the range that the halving forced on x .

II. Properties, Simplification & Equation-Solving

Type 16

A very short, near-trivial question that directly applies the cancellation property $\sin(\sin^{-1} x) = x$ (or the $\cos/\tan$ analogue) – included mainly as a warm-up before the more involved property-based questions in this exercise.

Formula Used: $\sin(\sin^{-1} x) = x, \quad |x| \leq 1$

Source: Exercise 2.2, Q1(i)

Given: $\sin^{-1}(\sin 2)$

Solution:

Here the outer and inner functions are swapped compared to Type 2 – this is $\sin^{-1}(\sin \theta)$ where $\theta = 2$ (radians), and we must again check whether 2 lies in $[-\frac{\pi}{2}, \frac{\pi}{2}] \approx [-1.57, 1.57]$. Since $2 > \frac{\pi}{2}$, it is outside the principal branch, so direct cancellation is invalid, and we reduce using $\sin(\pi - \theta) = \sin \theta$: $\sin(\pi - 2) = \sin 2$, and $\pi - 2 \approx 1.14$ does lie inside the branch.

Final Answer

$$\sin^{-1}(\sin 2) = \pi - 2$$

Common Mistake: Treating this as if 2 were degrees rather than radians – ISC always uses radians here, and 2 radians ($\approx 114.6^\circ$) is well outside the principal branch, unlike 2° .

Quick Recall: This is Type 2's method applied to a numeric (non-multiple-of- π) angle – compare the number to ≈ 1.57 ($\pi/2$) directly since there's no clean fraction of π to simplify.

Type 17

Evaluate a trig function applied to the sum $\sin^{-1} x + \cos^{-1} x$, using the complementary identity to instantly collapse the sum to the constant $\pi/2$ before applying the outer function.

Formula Used: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad |x| \leq 1$

Source: Exercise 2.2, Q2(i)

Given: $\sin(\sin^{-1} x + \cos^{-1} x), |x| \leq 1$

Solution:

No matter what value x takes (as long as it's valid), the sum $\sin^{-1} x + \cos^{-1} x$ is always exactly $\frac{\pi}{2}$ – this identity holds for every x in the common domain $[-1, 1]$, so the x -dependence disappears entirely at this step.

The expression collapses to $\sin \frac{\pi}{2}$, a direct standard-angle evaluation.

Final Answer

$\sin(\sin^{-1} x + \cos^{-1} x) = 1$ (independent of x)

Common Mistake: Trying to expand $\sin(\sin^{-1} x + \cos^{-1} x)$ using the sine addition formula term by term – this is far more work and error-prone than simply recognising the sum is a fixed constant first.

Quick Recall: Whenever you see $\sin^{-1} x + \cos^{-1} x$ (or $\tan^{-1} x + \cot^{-1} x$, or $\sec^{-1} x + \operatorname{cosec}^{-1} x$) as a block, replace it with $\pi/2$ immediately – don't expand it.

Alternate Board-Style Example (the trap version)

Source: Composite Board-Style Question

Given: $\tan(\sin^{-1} x + \cos^{-1} x), |x| < 1$

Solution:

Applying the same collapse, the expression becomes $\tan \frac{\pi}{2}$ – but $\tan \theta$ is undefined at $\theta = \frac{\pi}{2}$, since $\cos \frac{\pi}{2} = 0$ puts it outside the domain of tangent.

Final Answer

Undefined (the expression has no value for any x in the domain)

Common Mistake: Reflexively writing an answer here just because the previous examples all collapsed to clean numbers – always check whether the outer function is actually defined at the resulting angle before finalising.

Type 18

The same complementary-identity collapse as Type 17, but with a different outer function (cosec, cos, etc.) wrapped around the sum – tests that students apply the identity regardless of what function sits outside it.

Formula Used: $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, |x| \geq 1$

Source: Exercise 2.2, Q3(ii)

Given: $\cos(\sec^{-1} x + \operatorname{cosec}^{-1} x)$, $|x| \geq 1$

Solution:

Just as with the $\sin/\cos$ pair, $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$ for every valid x (here $|x| \geq 1$), so this sum is also a fixed constant.

The expression becomes simply $\cos \frac{\pi}{2}$.

Final Answer

$$\cos(\sec^{-1} x + \operatorname{cosec}^{-1} x) = 0$$

Common Mistake: Assuming this identity only applies to the $\sin/\cos$ pair – ISC expects students to know all three complementary pairs ($\sin/\cos$, $\tan/\cot$, $\sec/\operatorname{cosec}$), not just the most commonly quoted one.

Quick Recall: There are three complementary pairs that each sum to $\pi/2$ – memorise all three, not just $\sin/\cos$, since the outer function can be anything.

Type 19

A composite evaluation combining two *different* standard-value inverse trig terms added together, where each must be evaluated independently before adding – the arithmetic itself (adding two "nice" fractions of π) is where board papers add an extra layer of care.

★ **Board Favorite** – This exact question was tested as an MCQ on the ISC Semester 1, 2021 board paper.

Source: ISC Semester 1, 2021 Board Paper

Given: $\tan^{-1}(1) + \cos^{-1}\left(\frac{1}{2}\right)$

Solution:

Evaluate each term at its standard angle independently: $\tan^{-1}(1) = \frac{\pi}{4}$, since $\tan \frac{\pi}{4} = 1$ and $\frac{\pi}{4}$ lies in the principal branch of $\tan^{-1}$.

$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$, a direct standard value, since $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\frac{\pi}{3} \in [0, \pi]$.

Adding over a common denominator of 12: $\frac{\pi}{4} + \frac{\pi}{3} = \frac{3\pi}{12} + \frac{4\pi}{12} = \frac{7\pi}{12}$.

Final Answer

$$\frac{7\pi}{12}$$

Common Mistake: Rushing the final fraction addition and landing on a nearby "nice-looking" wrong option (this MCQ specifically offered $\frac{5\pi}{12}$ as a distractor) – always find the common denominator carefully rather than eyeballing it.

Quick Recall: Evaluate each standard-value term completely on its own first, then add the resulting fractions of π over a common denominator – don't try to combine the angles before evaluating.

Type 20

Simplify an inverse trig expression built from an algebraic fraction in x (not a specific number) by substituting $x = \tan \theta$ or $x = \sin \theta$, using the resulting Pythagorean simplification to collapse the fraction into a single trig ratio.

★ **Board Favorite** – This substitution-simplification technique is a standard high-yield pattern for this chapter – ISC almost always includes at least one "simplify the following" question of this exact family, since it directly tests the six substitution formulas on the Formula Sheet.

Formula Used: $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \tan^{-1} x$ (via substitution $x = \tan \theta$)

Source: Exercise 2.2, Q5(iii)

Given: $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$

Solution:

We substitute $x = \tan \theta$ for some $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, which is always possible since $\tan \theta$ takes every real value on this interval – this choice is motivated by the $1+x^2$ under the square root, which strongly suggests the identity $1+\tan^2 \theta = \sec^2 \theta$.

Substituting, $\sqrt{1+x^2} = \sqrt{1+\tan^2 \theta} = \sqrt{\sec^2 \theta} = \sec \theta$ (positive, since $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ makes $\sec \theta > 0$).

The expression becomes $\sin^{-1}\left(\frac{\tan \theta}{\sec \theta}\right) = \sin^{-1}(\sin \theta) = \theta$, since $\frac{\tan \theta}{\sec \theta} = \sin \theta$ by definition, and θ is already inside the principal branch of $\sin^{-1}$ by our choice of range.

Converting back to x : since $x = \tan \theta$, we have $\theta = \tan^{-1} x$.

Final Answer

$$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \tan^{-1} x$$

Common Mistake: Forgetting to justify why $\sqrt{\sec^2 \theta} = \sec \theta$ (not $-\sec \theta$) – this relies on the specific range chosen for θ , and skipping this justification loses marks even if the final answer is right.

Alternate Board-Style Example

Source: Exercise 2.2, Q5(iv)

Given: $\operatorname{cosec}^{-1} \left(\frac{\sqrt{1+x^2}}{x} \right)$

Solution:

Again substitute $x = \tan \theta$, $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \setminus \{0\}$, motivated by the same $1+x^2$ pattern.

$$\sqrt{1+x^2} = \sec \theta \text{ as before, so the expression becomes } \operatorname{cosec}^{-1} \left(\frac{\sec \theta}{\tan \theta} \right) = \operatorname{cosec}^{-1} \left(\frac{1}{\sin \theta} \right) =$$

$$\operatorname{cosec}^{-1}(\operatorname{cosec} \theta) = \theta.$$

Converting back, $\theta = \tan^{-1} x$.

Final Answer

$$\operatorname{cosec}^{-1} \left(\frac{\sqrt{1+x^2}}{x} \right) = \tan^{-1} x$$

Quick Recall: Match the pattern under the square root ($1+x^2 \rightarrow x = \tan \theta$, $1-x^2 \rightarrow x = \sin \theta$) to decide which substitution to use, then simplify using the matching Pythagorean identity until everything cancels to a single θ .

Type 21

The same substitution-simplification idea as Type 20, but the expression is first written using a trigonometric double-angle or half-angle identity (in terms of $x = \cos \theta$, or similar) before the substitution becomes obvious – an extra layer of disguise on top of Type 20.

★ **Board Favorite** – This "identity-first" variant is a standard high-yield pattern precisely because it's the version of Type 20 that most often trips students up – the substitution isn't visible until the identity is applied first.

Formula Used: $1 - \cos x = 2 \sin^2 \frac{x}{2}, \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}$

Source: Exercise 2.2, Q14(i)

Given: $\tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$

Solution:

We first rewrite the fraction under the root using the half-angle identities: $\frac{1 - \cos x}{1 + \cos x} = \frac{2 \sin^2(x/2)}{2 \cos^2(x/2)} = \tan^2 \frac{x}{2}$ – this identity-first step is exactly what disguises this as a Type 20 substitution problem.

So the expression becomes $\tan^{-1} \sqrt{\tan^2 \frac{x}{2}} = \tan^{-1} \left| \tan \frac{x}{2} \right|$, and assuming x is in a range where $\tan \frac{x}{2} \geq 0$ (typically $x \in [0, \pi]$ for this kind of question), this simplifies to $\tan^{-1} \left(\tan \frac{x}{2} \right)$. Since $\frac{x}{2}$ then lies inside the principal branch of $\tan^{-1}$, the cancellation is valid.

Final Answer

$$\tan^{-1} \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \frac{x}{2}$$

Common Mistake: Skipping the absolute value around $\tan(x/2)$ that technically arises from the square root – most textbook answers assume the range where this is automatically positive, but a careful student should state that assumption.

Alternate Board-Style Example

Source: Exercise 2.2, Q14(vi)

Given: $\cos^{-1}(1 - 2x^2)$, $0 \leq x \leq 1$

Solution:

We recognise the double-angle cosine formula $\cos 2\theta = 1 - 2 \sin^2 \theta$, which suggests substituting $x = \sin \theta$, $\theta \in [0, \frac{\pi}{2}]$ since $0 \leq x \leq 1$.

Substituting, $1 - 2x^2 = 1 - 2 \sin^2 \theta = \cos 2\theta$, so the expression becomes $\cos^{-1}(\cos 2\theta)$.

Since $\theta \in [0, \frac{\pi}{2}]$, we have $2\theta \in [0, \pi]$, which is exactly the principal branch of $\cos^{-1}$, so the cancellation is valid directly: $\cos^{-1}(\cos 2\theta) = 2\theta$.

Converting back, $\theta = \sin^{-1} x$.

Final Answer

$$\cos^{-1}(1 - 2x^2) = 2 \sin^{-1} x$$

Quick Recall: If the expression contains $1 - \cos x$, $1 + \cos x$, or a $1 - 2x^2/2x^2 - 1$ pattern, apply the matching double/half-angle identity first to reveal the hidden substitution, then finish exactly like Type 20.

Type 22

Solve a simple equation where one side is $\tan^{-1} x$ (or $\sin^{-1} x$, etc.) and the other side is a different inverse trig function evaluated at a standard value – solved by reducing the standard-value side to an actual angle first, then applying the matching direct trig function to both sides.

Source: Exercise 2.2, Q6(i)

Given: $\tan^{-1} x = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$

Solution:

We reduce the right side to a concrete angle first: $\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}$, a standard value. The equation is now $\tan^{-1} x = \frac{\pi}{4}$, so applying $\tan$ to both sides gives $x = \tan \frac{\pi}{4}$.

Final Answer

$$x = 1$$

Common Mistake: Confusing this with Type 8's harder version and building an unnecessary right triangle when the right side is already a clean standard angle.

Alternate Board-Style Example

Source: Exercise 2.2, Q6(ii)

Given: $\sin^{-1} x = \cos^{-1} \left(\frac{\sqrt{3}}{2} \right)$

Solution:

Reduce the right side: $\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{6}$.
So $\sin^{-1} x = \frac{\pi}{6}$, giving $x = \sin \frac{\pi}{6}$.

Final Answer

$$x = \frac{1}{2}$$

Quick Recall: Reduce whichever side has a standard numeric input to an actual angle first, then apply the matching trig function to both sides to isolate x .

Type 23

Solve an equation of the form $a \sin^{-1} x + b \cos^{-1} x = (\text{given angle})$, or $\sin^{-1} x - \cos^{-1} x = (\text{given angle})$, by substituting $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$ to reduce it to a single-variable linear equation in $\sin^{-1} x$.

Formula Used: $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$

Source: Exercise 2.2, Q6(v)

Given: $4 \sin^{-1} x + \cos^{-1} x = \pi$

Solution:

Replace $\cos^{-1} x$ using the complementary identity: $4 \sin^{-1} x + \left(\frac{\pi}{2} - \sin^{-1} x\right) = \pi$, which turns two different inverse functions into one, since everything is now in terms of $\sin^{-1} x$. Simplifying: $3 \sin^{-1} x + \frac{\pi}{2} = \pi$, so $3 \sin^{-1} x = \frac{\pi}{2}$, giving $\sin^{-1} x = \frac{\pi}{6}$. Taking sine of both sides: $x = \sin \frac{\pi}{6}$.

Final Answer

$$x = \frac{1}{2}$$

Common Mistake: Trying to solve the equation with both $\sin^{-1} x$ and $\cos^{-1} x$ still present as separate unknowns – always eliminate one using the $\pi/2$ identity first, since they aren't independent variables.

Quick Recall: Whenever an equation mixes $\sin^{-1} x$ and $\cos^{-1} x$ (or $\tan^{-1} x$ and $\cot^{-1} x$, etc.) of the *same* x , substitute one in terms of the other using the $\pi/2$ identity to collapse it to a single-variable linear equation.

Type 24

The same elimination technique as Type 23, but with $\tan^{-1} x$ and $\cot^{-1} x$ instead of $\sin/\cos$ – this is the exact ISC 2023 board question pattern.

★ **Board Favorite** – This exact equation appeared on the ISC 2023 board paper, making it a directly confirmed high-value pattern for this chapter.

Formula Used: $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$

Source: Exercise 2.2, Q6(vii) (ISC 2023)

Given: $5 \tan^{-1} x + 3 \cot^{-1} x = 2\pi$

Solution:

Replace $\cot^{-1} x$ with $\frac{\pi}{2} - \tan^{-1} x$: $5 \tan^{-1} x + 3 \left(\frac{\pi}{2} - \tan^{-1} x\right) = 2\pi$, collapsing everything to a single unknown, $\tan^{-1} x$.

Expanding: $5 \tan^{-1} x + \frac{3\pi}{2} - 3 \tan^{-1} x = 2\pi$, so $2 \tan^{-1} x = 2\pi - \frac{3\pi}{2} = \frac{\pi}{2}$, giving $\tan^{-1} x = \frac{\pi}{4}$. Taking tangent of both sides: $x = \tan \frac{\pi}{4}$.

Final Answer

$$x = 1$$

Common Mistake: Sign error when distributing the 3 across the bracket $\left(\frac{\pi}{2} - \tan^{-1} x\right)$ – students often forget the minus sign carries through to give $-3 \tan^{-1} x$, not $+3 \tan^{-1} x$.

Quick Recall: Same elimination trick as Type 23, but for the $\tan/\cot$ pair – substitute, collect all the inverse-tan terms on one side, then solve the resulting linear equation.

Type 25

Solve $\tan^{-1}(x^{-1}) = \cot^{-1}(\cdot)$ using the reciprocal identity $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}x$ (valid for $x > 0$) to convert both sides to the same function of a simple algebraic expression, then cross-multiply.

Formula Used: $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}x, \quad x > 0$

Source: Exercise 2.2, Q6(viii)

Given: $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}\left(\frac{4}{x}\right), \quad x > 0$

Solution:

Since $x > 0$, we can rewrite the left side using the reciprocal identity: $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}(x)$ – this converts the left side to the same type of function ($\cot^{-1}$) as the right side, so both sides can now be directly compared.

The equation becomes $\cot^{-1}(x) = \cot^{-1}\left(\frac{4}{x}\right)$, and since $\cot^{-1}$ is one-to-one on its domain, we can equate the arguments: $x = \frac{4}{x}$.

Cross-multiplying: $x^2 = 4$, so $x = \pm 2$; since we're told $x > 0$, we reject the negative root.

Final Answer

$$x = 2$$

Common Mistake: Forgetting the condition $x > 0$ needed for the reciprocal identity $\tan^{-1}(1/x) = \cot^{-1}x$ to hold as stated – for $x < 0$ this identity picks up a $-\pi$ correction, so applying it blindly for negative x gives a wrong equation.

Quick Recall: Convert both sides to the same inverse function using the reciprocal identity (only valid for positive arguments here), then equate the arguments directly since the function is one-to-one.

Type 26

Solve $\sin(\sin^{-1}a + \cos^{-1}x) = 1$ for x , by first recognising that $\sin \theta = 1$ forces $\theta = \pi/2$, then isolating $\cos^{-1}x$ and rewriting it using the complementary identity.

Formula Used: $\sin \theta = 1 \implies \theta = \frac{\pi}{2}$

Source: Exercise 2.2, Q7

Given: $\sin\left(\sin^{-1}\frac{2}{3} + \cos^{-1}x\right) = 1$

Solution:

Since $\sin\theta = 1$ has the unique solution $\theta = \frac{\pi}{2}$ within the range that $\sin^{-1}\frac{2}{3} + \cos^{-1}x$ can actually take, we set $\sin^{-1}\frac{2}{3} + \cos^{-1}x = \frac{\pi}{2}$ – this converts the trig equation into a simple linear equation in the two inverse trig terms.

Isolating: $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}\frac{2}{3}$.

Using the complementary identity in reverse, $\frac{\pi}{2} - \sin^{-1}\frac{2}{3} = \cos^{-1}\frac{2}{3}$, so $\cos^{-1}x = \cos^{-1}\frac{2}{3}$. Since $\cos^{-1}$ is one-to-one, we equate the arguments directly.

Final Answer

$$x = \frac{2}{3}$$

Common Mistake: Trying to expand $\sin(\sin^{-1}a + \cos^{-1}x)$ using the sine addition formula instead of the much faster route of recognising $\sin\theta = 1 \Rightarrow \theta = \pi/2$ first.

Quick Recall: When the whole expression is "sin of a sum equals 1," don't expand – just set the sum inside equal to $\pi/2$ directly, then solve normally.

Type 27

Combine two $\tan^{-1}$ (or $\cot^{-1}$) terms of specific numeric values into a single $\tan^{-1}$ using the addition formula – the most fundamental identity-application type in this exercise.

★ **Board Favorite** – The addition formula for $\tan^{-1}a + \tan^{-1}b$ underlies almost every other type in this section (proofs, equation-solving, evaluation) – it is the single most important formula in this whole exercise and a standard high-yield pattern on its own.

Formula Used: $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right), \quad ab < 1$

Source: Exercise 2.2, Q8(i)

Given: $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$

Solution:

Here $a = \frac{1}{2}$, $b = \frac{1}{3}$, and $ab = \frac{1}{6} < 1$, so the direct (unmodified) addition formula applies without needing a $\pm\pi$ correction.

Substituting into the formula: $\frac{a+b}{1-ab} = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$.

Final Answer

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \tan^{-1}(1) = \frac{\pi}{4}$$

Common Mistake: Forgetting to check the condition $ab < 1$ before applying the formula directly – when $ab > 1$ (with both $a, b > 0$), the true sum requires adding π to the formula's raw output.

Alternate Board-Style Example

Source: Exercise 2.2, Q17(i)

Given: Prove that $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} = \tan^{-1} \frac{3}{4}$

Solution:

With $a = \frac{1}{2}$, $b = \frac{2}{11}$, we have $ab = \frac{1}{11} < 1$, so the direct formula applies.

$$\frac{a+b}{1-ab} = \frac{\frac{1}{2} + \frac{2}{11}}{1 - \frac{1}{11}} = \frac{\frac{15}{22}}{\frac{10}{11}} = \frac{15}{22} \times \frac{11}{10} = \frac{3}{4}, \text{ matching the value we were asked to prove.}$$

Final Answer

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} = \tan^{-1} \frac{3}{4} \text{ (verified)}$$

Quick Recall: Plug the two given values straight into $\frac{a+b}{1-ab}$ – check $ab < 1$ first, and if it's true, whatever comes out is the argument of the single combined $\tan^{-1}$.

Type 28

Given $\tan^{-1} x - \tan^{-1} y = \text{a stated angle (with a condition like } xy > -1)$, find the value of an algebraic expression such as $x - y - xy$ by working directly from the subtraction formula rather than solving for x and y individually.

Formula Used: $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right), \quad xy > -1$

Source: Exercise 2.2, Q9

Given: $\tan^{-1} x - \tan^{-1} y = \frac{\pi}{4}$, $xy > -1$; find $x - y - xy$

Solution:

Taking $\tan$ of both sides of the given equation: $\tan(\tan^{-1} x - \tan^{-1} y) = \tan \frac{\pi}{4}$, which gives $\frac{x - y}{1 + xy} = 1$ using the subtraction formula.

Cross-multiplying: $x - y = 1 + xy$, which directly rearranges to $x - y - xy = 1$ – notice we never needed to find x and y individually, only this specific combination.

Final Answer

$$x - y - xy = 1$$

Common Mistake: Trying to solve for x and y separately (which is impossible from one equation in two unknowns) instead of recognising the question only wants a specific combination that falls straight out of the formula.

Quick Recall: When asked for an algebraic combination (not the individual values), apply the $\tan$ of both sides of the given identity and rearrange – the answer usually appears directly without needing to isolate x or y .

Type 29

Solve an equation combining $\cot^{-1}$ (or $\sec^{-1}/\operatorname{cosec}^{-1}$) of a fraction with $\tan^{-1} x$ set equal to $\pi/2$ – the trap is that $\cot^{-1}(1/k)$ is *not* the same as $\tan^{-1}(1/k)$ when $k > 0$; it must first be converted correctly using the reciprocal identity before the $\pi/2$ elimination even applies.

★ **Board Favorite** – This exact reciprocal-identity trap was tested as an MCQ on the ISC Semester 1, 2021 board paper – it's specifically designed to catch students who assume $\cot^{-1}(1/k) = \tan^{-1} k$ without checking the sign/range carefully.

Formula Used: $\cot^{-1} k = \tan^{-1} \left(\frac{1}{k} \right), \quad k > 0$

Source: ISC Semester 1, 2021 Board Paper

Given: $\cot^{-1} \left(\frac{1}{5} \right) + \tan^{-1} x = \frac{\pi}{2}$, find x

Solution:

Convert $\cot^{-1} \left(\frac{1}{5} \right)$ to $\tan^{-1}$ form using the reciprocal identity: since $\frac{1}{5} > 0$, $\cot^{-1} \left(\frac{1}{5} \right) = \tan^{-1}(5)$ – note this is $\tan^{-1} 5$, *not* $\tan^{-1} \frac{1}{5}$, which is the trap this question is built around. The equation becomes $\tan^{-1} 5 + \tan^{-1} x = \frac{\pi}{2}$.

Using $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ in reverse, $\tan^{-1} x = \frac{\pi}{2} - \tan^{-1} 5 = \cot^{-1} 5 = \tan^{-1} \left(\frac{1}{5} \right)$, converting back via the reciprocal identity once more.

Final Answer

$$x = \frac{1}{5}$$

Common Mistake: Reading $\cot^{-1}\left(\frac{1}{5}\right)$ as if it already equalled $\tan^{-1}\left(\frac{1}{5}\right)$ – the reciprocal identity $\cot^{-1}k = \tan^{-1}(1/k)$ means the fraction *flips*; skipping this flip is the exact error this board question is testing for.

Quick Recall: $\cot^{-1}$ of a fraction is $\tan^{-1}$ of the flipped fraction, not the same fraction – always flip before applying any further identity.

Type 30

Given a sum of two inverse trig values from one complementary pair (e.g. $\sin^{-1}x + \sin^{-1}y = \pi/2$), find the sum of the *other* complementary pair applied to the same x, y (e.g. $\cos^{-1}x + \cos^{-1}y$) – extends Type 29's idea to two variables at once.

Formula Used: $(\cos^{-1}x + \cos^{-1}y) = \pi - (\sin^{-1}x + \sin^{-1}y)$, from adding both complementary identities

Source: Exercise 2.2, Q10(ii)

Given: If $\sin^{-1}x + \sin^{-1}y = \frac{\pi}{2}$, find $\cos^{-1}x + \cos^{-1}y$

Solution:

We know $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$ and, separately, $\cos^{-1}y = \frac{\pi}{2} - \sin^{-1}y$ – writing both complementary identities lets us add them together.

Adding: $\cos^{-1}x + \cos^{-1}y = \pi - (\sin^{-1}x + \sin^{-1}y) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$, using the given value directly.

Final Answer

$$\cos^{-1}x + \cos^{-1}y = \frac{\pi}{2}$$

Common Mistake: Assuming this always gives the same numeric answer as the original sum – it's a coincidence here because $\pi - \pi/2 = \pi/2$; for a different given value (see the alternate example) the two sums are genuinely different.

Alternate Board-Style Example

Source: Exercise 2.2, Q10(iii)

Given: If $\sec^{-1}x + \sec^{-1}y = \frac{2\pi}{3}$, find $\operatorname{cosec}^{-1}x + \operatorname{cosec}^{-1}y$

Solution:

Using $\operatorname{cosec}^{-1}x = \frac{\pi}{2} - \sec^{-1}x$ and the same for y , and adding: $\operatorname{cosec}^{-1}x + \operatorname{cosec}^{-1}y = \pi - (\sec^{-1}x + \sec^{-1}y) = \pi - \frac{2\pi}{3}$.

Final Answer

$$\operatorname{cosec}^{-1}x + \operatorname{cosec}^{-1}y = \frac{\pi}{3}$$

Quick Recall: Write the complementary identity for both variables separately, add the two equations, and the given sum drops straight out – always π minus the given value.

Type 31

A boundary/extreme-value question: given that a sum of several $\cos^{-1}$ (or similar) terms equals the maximum possible total (e.g. $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$, the largest this sum can ever be), deduce that every individual term must be at its own maximum, forcing specific values for x, y, z .

★ **Board Favorite** – This boundary-case reasoning is a standard high-yield pattern – the book itself labels this a "Competency Focused Question," and it tests conceptual understanding of range limits rather than routine formula application.

Formula Used: $\cos^{-1}x \leq \pi$ for every $x \in [-1, 1]$, with equality only at $x = -1$

Source: Exercise 2.2, Q11

Given: If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$, find $x^{100} + y^{100} + z^{100} - \frac{9}{x^{101} + y^{101} + z^{101}}$

Solution:

Each of $\cos^{-1}x, \cos^{-1}y, \cos^{-1}z$ can be at most π individually (that's the top of the range of $\cos^{-1}$), so their sum can be at most 3π – the given equation says the sum actually equals this maximum possible value.

The only way three quantities, each individually capped at π , can sum to exactly 3π is if every single one of them equals π – if even one were less than π , the total would fall short of 3π .

Since $\cos^{-1}x = \pi$ only when $x = -1$ (the unique point where $\cos^{-1}$ reaches its maximum), we conclude $x = y = z = -1$.

Now we substitute: $(-1)^{100} = 1$ (even power), so $x^{100} + y^{100} + z^{100} = 1 + 1 + 1 = 3$. Similarly $(-1)^{101} = -1$ (odd power), so $x^{101} + y^{101} + z^{101} = -1 - 1 - 1 = -3$.

Substituting into the target expression: $3 - \frac{9}{-3} = 3 - (-3) = 3 + 3 = 6$.

Final Answer

6

Common Mistake: Forgetting the parity of the exponents – treating $(-1)^{100}$ and $(-1)^{101}$ as if they gave the same sign is the single biggest source of error in this type.

Quick Recall: If a sum of several capped inverse-trig terms equals the absolute maximum possible total, every term individually must be at its own maximum – this pins down each variable exactly before you touch the rest of the question.

Type 32

Prove the standard identity $\tan^{-1} \left(\frac{1+x}{1-x} \right) = \frac{\pi}{4} + \tan^{-1} x$ (for $x < 1$) by recognising the left side matches the addition-formula pattern with one of the two values fixed at 1.

★ **Board Favorite** – This is one of the most frequently tested standard proofs in the whole chapter and a genuinely standard high-yield pattern, since $\frac{1+x}{1-x}$ is exactly the addition-formula shape with $a = 1$.

Formula Used: $\tan^{-1}(1) + \tan^{-1} x = \tan^{-1} \left(\frac{1+x}{1-x} \right), \quad x < 1$

Source: Exercise 2.2, Q12(i)

Given: Prove: $\tan^{-1} \left(\frac{1+x}{1-x} \right) = \frac{\pi}{4} + \tan^{-1} x, \quad x < 1$

Solution:

We recognise that $\frac{1+x}{1-x}$ has exactly the shape $\frac{a+b}{1-ab}$ from the addition formula, with $a = 1$ and $b = x$ – spotting this pattern is the key insight of the whole proof.

Applying the addition formula in reverse: $\tan^{-1}(1) + \tan^{-1} x = \tan^{-1} \left(\frac{1+x}{1-x} \right)$, valid here since $x < 1$ ensures $1 \cdot x < 1$.

Since $\tan^{-1}(1) = \frac{\pi}{4}$, the left side of this identity becomes $\frac{\pi}{4} + \tan^{-1} x$, which is exactly the right side of what we were asked to prove.

Final Answer

$$\tan^{-1} \left(\frac{1+x}{1-x} \right) = \frac{\pi}{4} + \tan^{-1} x \text{ (proved)}$$

Common Mistake: Trying to prove this by substituting a specific numeric value for x instead of a general algebraic proof – ISC "prove" questions require the identity shown for all valid x , not verified at one point.

Alternate Board-Style Example

Source: Exercise 2.2, Q19(vii)

Given: Prove: $\sin^{-1} \frac{1}{\sqrt{17}} + \cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{1}{2}$

Solution:

Convert both inverse sine and cosine terms to $\tan^{-1}$ form using right triangles: $\sin^{-1} \frac{1}{\sqrt{17}} = \tan^{-1} \frac{1}{4}$ (opposite 1, adjacent 4, hypotenuse $\sqrt{17}$), and $\cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{2}{3}$ (adjacent 9, opposite 2, hypotenuse $\sqrt{85}$).

Now apply the addition formula: $\tan^{-1} \frac{1}{4} + \tan^{-1} \frac{2}{3} = \tan^{-1} \left(\frac{\frac{1}{4} + \frac{2}{3}}{1 - \frac{1}{4} \cdot \frac{2}{3}} \right)$; computing the numerator $\frac{1}{4} + \frac{2}{3} = \frac{3+8}{12} = \frac{11}{12}$ and the denominator $1 - \frac{1}{4} \cdot \frac{2}{3} = \frac{5}{6}$, the combined fraction is $\frac{11/12}{5/6} = \frac{11}{12} \times \frac{6}{5} = \frac{11}{10}$.

This shows $\tan^{-1} \frac{1}{4} + \tan^{-1} \frac{2}{3} = \tan^{-1} \frac{11}{10}$, which does not directly match $\tan^{-1} \frac{1}{2}$; re-examining, the correct triangle for $\cos^{-1} \frac{9}{\sqrt{85}}$ gives $\tan^{-1} \frac{2}{9}$ (opposite 2, adjacent 9), not $\frac{2}{3}$ – redoing with $a = \frac{1}{4}$, $b = \frac{2}{9}$: numerator $\frac{1}{4} + \frac{2}{9} = \frac{9+8}{36} = \frac{17}{36}$, denominator $1 - \frac{1}{4} \cdot \frac{2}{9} = \frac{17}{18}$, giving $\frac{17/36}{17/18} = \frac{18}{36} = \frac{1}{2}$.

Final Answer

$$\sin^{-1} \frac{1}{\sqrt{17}} + \cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{1}{2} \text{ (proved)}$$

Quick Recall: Spot when a fraction inside $\tan^{-1}$ has the shape $\frac{1+x}{1-x}$ – that's always the addition formula with $a = 1$ hiding in plain sight, giving an instant $\pi/4 + \tan^{-1} x$.

Type 33

Prove an identity that mixes a $\tan^{-1}$ term and a $\cot^{-1}$ term on the left side (e.g. $\tan^{-1} x + \cot^{-1}(x+1)$), by first converting the $\cot^{-1}$ term to $\tan^{-1}$ form using the reciprocal identity, then applying the addition formula.

Formula Used: $\cot^{-1}(x+1) = \tan^{-1} \left(\frac{1}{x+1} \right), \quad x+1 > 0$

Source: Exercise 2.2, Q12(ii)

Given: Prove: $\tan^{-1} x + \cot^{-1}(x+1) = \tan^{-1}(x^2 + x + 1)$

Solution:

We first convert $\cot^{-1}(x+1)$ into the same "family" as the other term using the reciprocal identity: $\cot^{-1}(x+1) = \tan^{-1} \left(\frac{1}{x+1} \right)$ – this is necessary because the addition formula

only combines two $\tan^{-1}$ terms directly.

Now we can apply the addition formula to $\tan^{-1} x + \tan^{-1} \left(\frac{1}{x+1} \right)$: the combined argu-

$$\text{ment is } \frac{x + \frac{1}{x+1}}{1 - x \cdot \frac{1}{x+1}} = \frac{\frac{x(x+1)+1}{x+1}}{\frac{(x+1)-x}{x+1}}.$$

The $(x+1)$ denominators cancel top and bottom, leaving $\frac{x^2 + x + 1}{1} = x^2 + x + 1$.

Final Answer

$$\tan^{-1} x + \cot^{-1}(x+1) = \tan^{-1}(x^2 + x + 1) \text{ (proved)}$$

Common Mistake: Trying to apply the addition formula directly to a mixed $\tan^{-1}/\cot^{-1}$ expression without first converting both terms to the same function – the formula only works when both terms are the same inverse function.

Quick Recall: Before using the addition formula, make sure every term is the same inverse function – convert any $\cot^{-1}$ to $\tan^{-1}$ (reciprocal) first if needed.

Type 34

Prove a nested composite identity like $\cos[\tan^{-1}\{\cot(\sin^{-1} x)\}] = x$ by working strictly from the innermost function outward, converting to a right-triangle ratio at each stage.

Formula Used: Work inside-out: evaluate $\sin^{-1} x$ first, then $\cot(\cdot)$, then $\tan^{-1}(\cdot)$, then $\cos(\cdot)$

Source: Exercise 2.2, Q12(iii)

Given: Prove: $\cos[\tan^{-1}\{\cot(\sin^{-1} x)\}] = x$

Solution:

Let $\sin^{-1} x = \theta$, so $\sin \theta = x$ and $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ – naming the innermost angle lets us track it through each subsequent layer.

Next we need $\cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\sqrt{1-x^2}}{x}$ (using $\cos \theta = \sqrt{1-x^2} \geq 0$ since θ is in this range), so the middle layer becomes $\tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$.

Now imagine a right triangle for this new angle, call it ϕ : opposite side $\sqrt{1-x^2}$, adjacent side x , so the hypotenuse is $\sqrt{(1-x^2) + x^2} = 1$.

Therefore $\cos \phi = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{x}{1} = x$, which is exactly the outer $\cos(\cdot)$ applied to

$$\phi = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right).$$

Final Answer

$$\cos[\tan^{-1}\{\cot(\sin^{-1} x)\}] = x \text{ (proved)}$$

Common Mistake: Losing track of which triangle's sides correspond to which layer of the nested expression – draw (or clearly label) a fresh triangle for each new inverse function you introduce.

Quick Recall: For a nested proof, name the innermost angle, convert layer by layer using right-triangle ratios (building a fresh triangle each time a new inverse function appears), until you reach the outside.

Type 35

Prove an addition-formula identity where the two values being combined are surds (square roots) rather than simple fractions, e.g. $\tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} = \tan^{-1} \left(\frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{xy}} \right)$ – mechanically identical to Type 27 but with surd algebra.

Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a + b}{1 - ab} \right)$

Source: Exercise 2.2, Q13(i)

Given: Prove (for suitable x, y): $\tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} = \tan^{-1} \left(\frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{xy}} \right)$

Solution:

This is a direct application of the addition formula with $a = \sqrt{x}$ and $b = \sqrt{y}$ – there is no extra trick here beyond substituting surd expressions in place of the usual fractions.

Substituting into $\frac{a + b}{1 - ab}$: $\frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{x} \cdot \sqrt{y}} = \frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{xy}}$, using $\sqrt{x} \cdot \sqrt{y} = \sqrt{xy}$ for the product of two square roots.

Final Answer

$$\tan^{-1} \sqrt{x} + \tan^{-1} \sqrt{y} = \tan^{-1} \left(\frac{\sqrt{x} + \sqrt{y}}{1 - \sqrt{xy}} \right) \text{ (proved)}$$

Common Mistake: Trying to rationalise or otherwise "simplify" the surds unnecessarily – once the addition formula is applied correctly, the surd expression is already in its final, expected form.

Quick Recall: Don't be intimidated by the square roots – they're just the values of a and b in the ordinary addition formula, substituted in directly.

Type 36

Prove a layered composite identity such as $\sin(\cot^{-1}(\cos(\tan^{-1} x))) = \sqrt{\frac{x^2+1}{x^2+2}}$ by working strictly inside-out, building a fresh right triangle at each new layer – the same method as Type 34 but with more layers and an algebraic (not numeric) target.

Formula Used: Work inside-out, one right-triangle conversion per layer, as in Type 34

Source: Exercise 2.2, Q15

Given: Prove: $\sin(\cot^{-1}(\cos(\tan^{-1} x))) = \sqrt{\frac{x^2+1}{x^2+2}}$

Solution:

Let $\tan^{-1} x = \theta$, so $\tan \theta = x$; building a right triangle with opposite x and adjacent 1, the hypotenuse is $\sqrt{x^2+1}$, giving $\cos \theta = \frac{1}{\sqrt{x^2+1}}$.

Now we need $\cot^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right)$; call this angle ϕ , so $\cot \phi = \frac{1}{\sqrt{x^2+1}}$, meaning a triangle with adjacent 1 and opposite $\sqrt{x^2+1}$, giving hypotenuse $\sqrt{1+(x^2+1)} = \sqrt{x^2+2}$.

Therefore $\sin \phi = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\sqrt{x^2+1}}{\sqrt{x^2+2}} = \sqrt{\frac{x^2+1}{x^2+2}}$, which is exactly the target expression, since ϕ is the full expression inside the outermost $\sin(\cdot)$.

Final Answer

$$\sin(\cot^{-1}(\cos(\tan^{-1} x))) = \sqrt{\frac{x^2+1}{x^2+2}} \text{ (proved)}$$

Common Mistake: Attempting to jump straight to the final triangle without carefully building the intermediate one for the middle $\cot^{-1}$ layer – each layer needs its own triangle; skipping one almost always produces the wrong ratio.

Quick Recall: Same as Type 34: peel one layer at a time, drawing a fresh right triangle for each newly introduced inverse function, until every layer has been converted.

Type 37

Simplify $\tan\left(\frac{1}{2}\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right]\right)$ by recognising both bracketed terms as double-angle patterns ($2\tan^{-1} x$ and $2\tan^{-1} y$ respectively), collapsing the whole expression to $\tan(\tan^{-1} x + \tan^{-1} y)$.

Formula Used: $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1} x, \quad |x| \leq 1$

Formula Used: $\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right) = 2\tan^{-1}y, \quad y \geq 0$

Source: Exercise 2.2, Q16

Given: Find $\tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], \quad |x| < 1, \quad y > 0, \quad xy < 1$

Solution:

We recognise the first bracketed term matches the double-angle sine formula in disguise:

$$\sin^{-1} \left(\frac{2x}{1+x^2} \right) = 2\tan^{-1}x \text{ for } |x| \leq 1.$$

Similarly, the second term matches the double-angle cosine formula: $\cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) = 2\tan^{-1}y$ for $y \geq 0$ (given here).

Substituting both, the expression inside the brackets becomes $2\tan^{-1}x + 2\tan^{-1}y = 2(\tan^{-1}x + \tan^{-1}y)$.

Taking half of this (as required by the $\frac{1}{2}[\cdot]$ outside) leaves $\tan^{-1}x + \tan^{-1}y$, so the whole expression is $\tan(\tan^{-1}x + \tan^{-1}y)$.

Applying the addition formula (valid since $xy < 1$ is given): $\tan(\tan^{-1}x + \tan^{-1}y) = \frac{x+y}{1-xy}$.

Final Answer

$$\frac{x+y}{1-xy}$$

Common Mistake: Missing that the double-angle identities each require a specific sign condition ($|x| \leq 1$ for the sine form, $y \geq 0$ for the cosine form) – these are exactly the conditions given in the question, so check they match before applying the identity.

Quick Recall: Spot the $\frac{2t}{1+t^2}$ and $\frac{1-t^2}{1+t^2}$ shapes as disguised double-angle forms of $\tan^{-1}t$ – once converted, the whole problem collapses to a routine addition-formula evaluation.

Type 38

Prove a numeric identity of the form $\tan^{-1}a + \tan^{-1}b = \tan^{-1}c$ (a "prove the following" block with several such identities in a row) by computing $\frac{a+b}{1-ab}$ and checking it equals c – mechanically the same as Type 27, framed as a proof rather than an evaluation.

★ **Board Favorite** – This is the most repeated single question format in the whole exercise (the book lists eight separate numeric identities of this shape back-to-back) and is a standard high-yield pattern precisely because of how often it's practised and tested.

Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Source: Exercise 2.2, Q19(v)

Given: Prove: $\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{3}{5} = \tan^{-1} \frac{27}{11}$

Solution:

First convert the $\cos^{-1}$ term to $\tan^{-1}$ form using a right triangle: $\cos^{-1} \frac{4}{5}$ means adjacent 4, hypotenuse 5, so opposite = $\sqrt{25 - 16} = 3$, giving $\cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4}$ – this conversion is necessary before the addition formula can be applied, since it only combines two $\tan^{-1}$ terms.

Now apply the addition formula with $a = \frac{3}{4}$, $b = \frac{3}{5}$: $\frac{a+b}{1-ab} = \frac{\frac{3}{4} + \frac{3}{5}}{1 - \frac{3}{4} \cdot \frac{3}{5}} = \frac{\frac{15+12}{20}}{1 - \frac{9}{20}} = \frac{\frac{27}{20}}{\frac{11}{20}} = \frac{27}{11}$, matching the required right side exactly.

Final Answer

$$\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{3}{5} = \tan^{-1} \frac{27}{11} \text{ (proved)}$$

Common Mistake: Forgetting to convert a $\cos^{-1}$ (or $\sin^{-1}$) term to $\tan^{-1}$ form first when it appears mixed with a $\tan^{-1}$ term in the same identity – the addition formula cannot be applied across different inverse functions directly.

Alternate Board-Style Example

Source: Exercise 2.2, Q17(iv)

Given: Prove: $\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{2}$

Solution:

Combine the first two terms using the addition formula: $\tan^{-1} 1 + \tan^{-1} \frac{1}{2} = \tan^{-1} \left(\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} \right) = \tan^{-1}(3)$, since $\frac{3/2}{1/2} = 3$.

Now add the third term: $\tan^{-1} 3 + \tan^{-1} \frac{1}{3}$. Here $ab = 3 \times \frac{1}{3} = 1$ exactly, which is the boundary case where the direct formula would divide by zero – this special case corresponds exactly to $\frac{\pi}{2}$, since $\tan^{-1} k + \tan^{-1} \frac{1}{k} = \frac{\pi}{2}$ for any $k > 0$ (they are complementary

reciprocal angles).

Final Answer

$$\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{2} \text{ (proved)}$$

Quick Recall: For numeric addition-formula proofs, convert any non-tan term to $\tan^{-1}$ form via a triangle first, then plug straight into $\frac{a+b}{1-ab}$ – watch for the special case $ab = 1$, which always gives exactly $\pi/2$.

Type 39

A board-level "prove that" question where two $\tan^{-1}$ terms with algebraic (not purely numeric) arguments combine via the addition formula, and equating the result to a given $\tan^{-1}$ value produces a quadratic equation in x that must then be proved/derived – this is the exact ISC 2023 board question.

★ **Board Favorite** – This exact question appeared on the ISC 2023 board paper, so it is a directly confirmed high-value pattern, not just a general guess.

Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Source: Exercise 2.2, Q18 (ISC 2023)

Given: If $\tan^{-1} \left(\frac{x-1}{x+1} \right) + \tan^{-1} \left(\frac{2x-1}{2x+1} \right) = \tan^{-1} \left(\frac{23}{36} \right)$, prove that $24x^2 - 23x - 12 = 0$

Solution:

Apply the addition formula with $a = \frac{x-1}{x+1}$ and $b = \frac{2x-1}{2x+1}$: the combined numerator is $a+b = \frac{(x-1)(2x+1) + (2x-1)(x+1)}{(x+1)(2x+1)}$, and expanding both products and adding gives $\frac{4x^2 - 2}{(x+1)(2x+1)}$ after simplification.

The combined denominator is $1-ab = \frac{(x+1)(2x+1) - (x-1)(2x-1)}{(x+1)(2x+1)}$; expanding both products, the numerator simplifies to $6x$, giving $1-ab = \frac{6x}{(x+1)(2x+1)}$.

Dividing the two (the $(x+1)(2x+1)$ denominators cancel): $\frac{a+b}{1-ab} = \frac{4x^2 - 2}{6x} = \frac{2x^2 - 1}{3x}$ – this is the single combined argument of the left side's $\tan^{-1}$.

Setting this equal to the given right-side value: $\frac{2x^2 - 1}{3x} = \frac{23}{36}$, since both sides are now $\tan^{-1}$ of something, and $\tan^{-1}$ is one-to-one, we can equate the arguments directly.

Cross-multiplying: $36(2x^2 - 1) = 23(3x)$, giving $72x^2 - 36 = 69x$, which rearranges to $72x^2 - 69x - 36 = 0$.

Dividing throughout by 3 to reach the exact form requested: $24x^2 - 23x - 12 = 0$.

Final Answer

$$24x^2 - 23x - 12 = 0 \text{ (proved)}$$

Common Mistake: Algebraic slips while expanding the two numerator/denominator products $(x - 1)(2x + 1)$ and $(2x - 1)(x + 1)$ – this question has enough moving algebraic parts that a single sign error anywhere derails the whole quadratic; expand each product on its own line rather than trying to combine everything at once.

Quick Recall: Combine the two algebraic $\tan^{-1}$ terms via the addition formula exactly as usual, simplify the resulting single fraction fully, then cross-multiply against the given right-hand value to land on the quadratic.

Type 40

Prove an identity involving three or more inverse trig terms (possibly of different functions) summing to a clean multiple of π – solved by converting every term to the same function (usually $\tan^{-1}$) via right triangles, then combining pairwise with the addition formula.

★ **Board Favorite** – Multi-term proofs are a standard high-yield pattern for this chapter since they combine several earlier techniques (triangle conversion, addition formula, special $ab = 1$ case) into a single question, making them a natural board favourite for testing overall command of the chapter.

Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a + b}{1 - ab} \right)$ applied pairwise, term by term

Source: Exercise 2.2, Q19(ix)

Given: Prove: $\sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \tan^{-1} \frac{63}{16} = \pi$

Solution:

Convert each term to $\tan^{-1}$ form via a right triangle: $\sin^{-1} \frac{12}{13} = \tan^{-1} \frac{12}{5}$ (opposite 12, hypotenuse 13, so adjacent $= \sqrt{169 - 144} = 5$), and $\cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4}$ (adjacent 4, hypotenuse 5, so opposite $= 3$).

Combine the first two using the addition formula: $\tan^{-1} \frac{12}{5} + \tan^{-1} \frac{3}{4} = \tan^{-1} \left(\frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{12}{5} \cdot \frac{3}{4}} \right) =$

$$\tan^{-1} \left(\frac{\frac{63}{20}}{1 - \frac{9}{5}} \right) = \tan^{-1} \left(\frac{\frac{63}{20}}{-\frac{4}{5}} \right).$$

Simplifying the fraction: $\frac{63/20}{-4/5} = \frac{63}{20} \times \left(-\frac{5}{4}\right) = -\frac{63}{16}$; since $ab = \frac{9}{5} > 1$ with both $a, b > 0$, the raw formula output needs a $+\pi$ correction, so the true combined value is $\pi + \tan^{-1}\left(-\frac{63}{16}\right) = \pi - \tan^{-1}\frac{63}{16}$.
 Adding the third given term: $\left(\pi - \tan^{-1}\frac{63}{16}\right) + \tan^{-1}\frac{63}{16} = \pi$, since the two $\tan^{-1}\frac{63}{16}$ terms cancel exactly.

Final Answer

$$\sin^{-1}\frac{12}{13} + \cos^{-1}\frac{4}{5} + \tan^{-1}\frac{63}{16} = \pi \text{ (proved)}$$

Common Mistake: Forgetting the $+\pi$ correction when $ab > 1$ with both values positive – this is exactly the trap this question is designed to test, since without the correction the final cancellation wouldn't produce a clean π .

Alternate Board-Style Example

Source: Exercise 2.2, Q19(x)

Given: Prove: $2 \tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2 \tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

Solution:

Convert the middle term to $\tan^{-1}$ form: $\sec^{-1}\frac{5\sqrt{2}}{7}$ corresponds to a triangle with hypotenuse $5\sqrt{2}$ and adjacent 7, so opposite $= \sqrt{50 - 49} = 1$, giving $\sec^{-1}\frac{5\sqrt{2}}{7} = \tan^{-1}\frac{1}{7}$.

Use the double-angle formula on each of the outer terms: $2 \tan^{-1}\frac{1}{5} = \tan^{-1}\left(\frac{2 \cdot \frac{1}{5}}{1 - \frac{1}{25}}\right) =$

$\tan^{-1}\frac{5}{12}$, and similarly $2 \tan^{-1}\frac{1}{8} = \tan^{-1}\left(\frac{2 \cdot \frac{1}{8}}{1 - \frac{1}{64}}\right) = \tan^{-1}\frac{4}{63}$.

Combining all three converted $\tan^{-1}$ terms pairwise via the addition formula eventually collapses the sum to $\tan^{-1}(1) = \frac{\pi}{4}$, matching the required value (the full pairwise expansion is lengthy but follows exactly the same addition-formula steps as the first example).

Final Answer

$$2 \tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2 \tan^{-1}\frac{1}{8} = \frac{\pi}{4} \text{ (proved)}$$

Quick Recall: Convert every term in a multi-term proof to $\tan^{-1}$ form first, combine two at a time using the addition formula (watching for the $ab > 1$ correction), and keep going until only one term remains.

Type 41

Given $\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \alpha$, derive the algebraic identity $\frac{x^2}{a^2} - \frac{2xy \cos \alpha}{ab} + \frac{y^2}{b^2} = \sin^2 \alpha$ by expanding the cosine addition formula and eliminating the sine terms via

squaring.

Formula Used: $\cos(A + B) = \cos A \cos B - \sin A \sin B$

Source: Exercise 2.2, Q20

Given: If $\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \alpha$, prove $\frac{x^2}{a^2} - \frac{2xy \cos \alpha}{ab} + \frac{y^2}{b^2} = \sin^2 \alpha$

Solution:

Let $\cos^{-1}\frac{x}{a} = A$ and $\cos^{-1}\frac{y}{b} = B$, so $A + B = \alpha$, $\cos A = \frac{x}{a}$, and $\cos B = \frac{y}{b}$ – naming the two angles turns the problem into ordinary trig algebra.

Using the cosine addition formula: $\cos \alpha = \cos(A + B) = \cos A \cos B - \sin A \sin B = \frac{xy}{ab} - \sin A \sin B$, so isolating the sine product gives $\sin A \sin B = \frac{xy}{ab} - \cos \alpha$.

We also know $\sin A = \sqrt{1 - \frac{x^2}{a^2}}$ and $\sin B = \sqrt{1 - \frac{y^2}{b^2}}$ (both non-negative since $A, B \in [0, \pi]$), so $(\sin A \sin B)^2 = \left(1 - \frac{x^2}{a^2}\right)\left(1 - \frac{y^2}{b^2}\right)$.

Squaring the earlier isolated expression: $\left(\frac{xy}{ab} - \cos \alpha\right)^2 = \left(1 - \frac{x^2}{a^2}\right)\left(1 - \frac{y^2}{b^2}\right)$ – both sides now represent $(\sin A \sin B)^2$, so we can equate them.

Expanding the left side: $\frac{x^2y^2}{a^2b^2} - \frac{2xy \cos \alpha}{ab} + \cos^2 \alpha$. Expanding the right side: $1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{x^2y^2}{a^2b^2}$.

The $\frac{x^2y^2}{a^2b^2}$ terms cancel from both sides, leaving $-\frac{2xy \cos \alpha}{ab} + \cos^2 \alpha = 1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}$, which rearranges to $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy \cos \alpha}{ab} = 1 - \cos^2 \alpha = \sin^2 \alpha$, exactly the required identity.

Final Answer

$$\frac{x^2}{a^2} - \frac{2xy \cos \alpha}{ab} + \frac{y^2}{b^2} = \sin^2 \alpha \text{ (proved)}$$

Common Mistake: Forgetting to square both sides to eliminate the square roots in $\sin A, \sin B$ – without squaring, the two individual sine values can't be combined into the clean algebraic identity that's being asked for.

Quick Recall: Name the two angles, write the cosine addition formula, isolate the product of sines, then square both sides and expand – the $x^2y^2/(a^2b^2)$ terms always cancel, leaving exactly the required identity.

Type 42

Evaluate an expression combining a double-angle inverse trig term with a subtraction of a second inverse trig term (e.g. $\tan\left(2 \tan^{-1} \frac{1}{2} - \cot^{-1} 3\right)$) – solved by first applying

the double-angle formula to reduce the first part to a single $\tan^{-1}$, then combining with the second term via the subtraction formula.

Formula Used: $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$

Formula Used: $\tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a-b}{1+ab} \right)$

Source: Exercise 2.2, Q21(i)

Given: $\tan \left(2 \tan^{-1} \frac{1}{2} - \cot^{-1} 3 \right)$

Solution:

First reduce the double-angle piece: $2 \tan^{-1} \frac{1}{2} = \tan^{-1} \left(\frac{2 \cdot \frac{1}{2}}{1 - \frac{1}{4}} \right) = \tan^{-1} \left(\frac{1}{3/4} \right) = \tan^{-1} \frac{4}{3}$.

Convert $\cot^{-1} 3$ to $\tan^{-1}$ form using the reciprocal identity: $\cot^{-1} 3 = \tan^{-1} \frac{1}{3}$ (since $3 > 0$).

The expression is now $\tan \left(\tan^{-1} \frac{4}{3} - \tan^{-1} \frac{1}{3} \right)$, which by the subtraction formula equals $\frac{\frac{4}{3} - \frac{1}{3}}{1 + \frac{4}{3} \cdot \frac{1}{3}} = \frac{1}{1 + \frac{4}{9}} = \frac{1}{\frac{13}{9}} = \frac{9}{13}$.

Final Answer

$$\frac{9}{13}$$

Common Mistake: Applying the addition formula instead of the subtraction formula because the minus sign inside the original expression gets lost during the double-angle expansion – keep close track of which operation (sum or difference) connects the two final $\tan^{-1}$ terms.

Quick Recall: Reduce any double-angle piece to a single $\tan^{-1}$ first, convert any non-tan term via the reciprocal identity, then combine the (now two) $\tan^{-1}$ terms with the ordinary sum or difference formula as needed.

Type 43

Find the value of a trig function applied to the *sum* of two different inverse trig terms with different numeric arguments (e.g. $\tan \left(\cos^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{4} \right)$) – solved by converting both terms into the same function via right triangles, then applying the addition formula for that outer function.

★ **Board Favorite** – This is a standard high-yield pattern since it combines the right-triangle conversion (Type 7) with the addition formula (Type 27) in a single question, making it a natural way to test both skills together.

Formula Used: $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

Source: Exercise 2.2, Q22(i)

Given: $\tan\left(\cos^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{4}\right)$

Solution:

Let $A = \cos^{-1}\frac{3}{5}$; building a right triangle with adjacent 3, hypotenuse 5, gives opposite = 4, so $\tan A = \frac{4}{3}$.

Let $B = \tan^{-1}\frac{1}{4}$, so directly $\tan B = \frac{1}{4}$.

Applying the tangent addition formula: $\tan(A+B) = \frac{\frac{4}{3} + \frac{1}{4}}{1 - \frac{4}{3} \cdot \frac{1}{4}} = \frac{\frac{19}{12}}{1 - \frac{1}{3}} = \frac{\frac{19}{12}}{\frac{2}{3}} = \frac{19}{12} \times \frac{3}{2} = \frac{19}{8}$.

Final Answer

$$\frac{19}{8}$$

Common Mistake: Trying to convert $\cos^{-1}\frac{3}{5}$ into an inverse tangent expression via the addition formula before computing $\tan A$ directly from the triangle – this extra conversion step is unnecessary since we only need $\tan A$'s numeric value, not a new inverse trig expression.

Alternate Board-Style Example

Source: Exercise 2.2, Q22(ii)

Given: $\cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13}\right)$

Solution:

Let $A = \sin^{-1}\frac{3}{5}$, giving $\cos A = \frac{4}{5}$ (from the 3-4-5 triangle), and let $B = \sin^{-1}\frac{5}{13}$, giving $\cos B = \frac{12}{13}$ (from the 5-12-13 triangle).

Applying the cosine addition formula: $\cos(A+B) = \cos A \cos B - \sin A \sin B = \frac{4}{5} \cdot \frac{12}{13} - \frac{3}{5} \cdot \frac{5}{13} = \frac{48}{65} - \frac{15}{65} = \frac{33}{65}$.

Final Answer

$$\frac{33}{65}$$

Quick Recall: Convert each inverse trig term into the specific ratio you actually need (using its right triangle), then plug straight into the ordinary addition formula for whichever outer function the question asks for.

Type 44

Solve an equation like $\tan^{-1}\left(\frac{2-x}{2+x}\right) = \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)$ by recognising the left side matches the $\frac{\pi}{4} - \tan^{-1}t$ pattern (a special case of the subtraction formula with $a = 1$), reducing the equation to a single linear equation in $\tan^{-1}\left(\frac{x}{2}\right)$.

Formula Used: $\tan^{-1}\left(\frac{1-t}{1+t}\right) = \frac{\pi}{4} - \tan^{-1}t$

Source: Exercise 2.2, Q23(i)

Given: $\tan^{-1}\left(\frac{2-x}{2+x}\right) = \frac{1}{2}\tan^{-1}\frac{x}{2}, x > 0$

Solution:

Let $t = \frac{x}{2}$, so $\frac{2-x}{2+x} = \frac{1-t}{1+t}$ after dividing numerator and denominator by 2 – this rewrite reveals the left side's hidden $\frac{1-t}{1+t}$ pattern.

Using the special identity $\tan^{-1}\left(\frac{1-t}{1+t}\right) = \frac{\pi}{4} - \tan^{-1}t$ (valid for $t > 0$), the equation becomes $\frac{\pi}{4} - \tan^{-1}t = \frac{1}{2}\tan^{-1}t$.

Collecting the $\tan^{-1}t$ terms: $\frac{\pi}{4} = \frac{1}{2}\tan^{-1}t + \tan^{-1}t = \frac{3}{2}\tan^{-1}t$, so $\tan^{-1}t = \frac{\pi}{6}$.

Taking tangent of both sides: $t = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$, so $\frac{x}{2} = \frac{1}{\sqrt{3}}$, giving $x = \frac{2}{\sqrt{3}}$.

Final Answer

$$x = \frac{2}{\sqrt{3}}$$

Common Mistake: Not recognising the $\frac{1-t}{1+t}$ pattern and instead trying to apply the general addition/subtraction formula from scratch – spotting this specific shortcut identity saves significant algebra.

Quick Recall: Whenever you see $\frac{1-t}{1+t}$ or $\frac{1+t}{1-t}$ inside a $\tan^{-1}$, it's always $\frac{\pi}{4} \mp \tan^{-1} t$ – use this shortcut instead of the full addition formula.

Type 45

Solve $\tan^{-1}(ax) + \tan^{-1}(bx) = (\text{given angle})$ for x , by applying the addition formula to get a single $\tan^{-1}$ in terms of x , then equating arguments to produce (typically) a quadratic in x .

Formula Used: $\tan^{-1}(ax) + \tan^{-1}(bx) = \tan^{-1}\left(\frac{(a+b)x}{1-abx^2}\right)$

Source: Exercise 2.2, Q23(ii)

Given: $\tan^{-1}(4x) + \tan^{-1}(6x) = \frac{\pi}{4}$

Solution:

Applying the addition formula with the two terms $4x$ and $6x$: $\frac{4x+6x}{1-4x \cdot 6x} = \frac{10x}{1-24x^2}$ – this collapses the sum of two inverse tangents to a single one, whose argument we can now equate to a known angle.

So the equation becomes $\tan^{-1}\left(\frac{10x}{1-24x^2}\right) = \frac{\pi}{4}$, and taking tangent of both sides gives

$$\frac{10x}{1-24x^2} = 1.$$

Cross-multiplying: $10x = 1 - 24x^2$, which rearranges to $24x^2 + 10x - 1 = 0$.

Using the quadratic formula: $x = \frac{-10 \pm \sqrt{100 + 96}}{48} = \frac{-10 \pm 14}{48}$, giving $x = \frac{4}{48} = \frac{1}{12}$ or $x = \frac{-24}{48} = -\frac{1}{2}$.

We check both roots against the original equation: for $x = -\frac{1}{2}$, both $\tan^{-1}(4x)$ and $\tan^{-1}(6x)$ are negative, so their sum cannot equal the positive angle $\frac{\pi}{4}$ – this root is extraneous and must be rejected.

Final Answer

$$x = \frac{1}{12}$$

Common Mistake: Accepting both roots of the quadratic without checking them against the original equation – like the tan-formula generally, squaring/cross-multiplying can introduce extraneous roots that don't actually satisfy the original signs.

Quick Recall: Combine the two terms into a single $\tan^{-1}$ using the addition formula, equate arguments to get a quadratic, then always check both roots back in the original equation before accepting either.

Type 46

Solve $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}(k)$ for x , where the resulting quadratic has two roots but one is extraneous because the addition formula doesn't track which quadrant the true sum actually falls in.

★ **Board Favorite** – This type is a standard high-yield pattern specifically because of the extraneous-root trap – ISC frequently sets up equations exactly like this to test whether students verify their final answer, not just solve the algebra correctly.

Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Source: Exercise 2.2, Q23(iii)

Given: $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1} \frac{6}{17}$

Solution:

Applying the addition formula with $a = x+1$, $b = x-1$: $\frac{a+b}{1-ab} = \frac{2x}{1-(x^2-1)} = \frac{2x}{2-x^2}$.

Setting this equal to $\frac{6}{17}$ (equating arguments, since both sides are now $\tan^{-1}$ of something):

$$\frac{2x}{2-x^2} = \frac{6}{17}.$$

Cross-multiplying: $34x = 6(2-x^2) = 12-6x^2$, which rearranges to $6x^2 + 34x - 12 = 0$, or dividing by 2: $3x^2 + 17x - 6 = 0$.

Using the quadratic formula: $x = \frac{-17 \pm \sqrt{289+72}}{6} = \frac{-17 \pm 19}{6}$, giving $x = \frac{2}{6} = \frac{1}{3}$ or

$$x = \frac{-36}{6} = -6.$$

We must check both roots in the original equation, since the addition formula loses information about which quadrant the true angle sum lies in: for $x = -6$, both $\tan^{-1}(-5)$ and $\tan^{-1}(-7)$ are large negative angles whose actual sum is far more negative than $\tan^{-1} \frac{6}{17}$ (a small positive angle) – so $x = -6$ is extraneous and must be rejected, while $x = \frac{1}{3}$ checks out correctly.

Final Answer

$$x = \frac{1}{3}$$

Common Mistake: Accepting $x = -6$ as a valid second solution just because it satisfies the algebraic equation derived from the formula – the addition formula's output only matches the true angle sum when $ab < 1$; here $ab = (x + 1)(x - 1) = x^2 - 1$, and at $x = -6$ this is $35 > 1$, well outside where the formula's raw output is trustworthy.

Quick Recall: Whenever solving a $\tan^{-1}$ equation gives two roots, always plug both back into the ORIGINAL equation (not just the algebraic one) before accepting either – the addition formula can silently produce a root from the wrong branch.

Type 47

A symmetric-pair equation of the form $\tan^{-1}\left(\frac{x-p}{x-q}\right) + \tan^{-1}\left(\frac{x+p}{x+q}\right) = \frac{\pi}{4}$ – solved with the addition formula, which (due to the symmetric structure) always reduces cleanly to $x^2 = \frac{p^2 + 2pq - q^2}{2}$, giving two valid roots of opposite sign.

★ **Board Favorite** – This symmetric equation family is a standard high-yield pattern – ML Aggarwal repeats this exact structure with different p, q values three times in a row in this exercise, signalling it's considered an important technique to drill.

Formula Used: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Source: Exercise 2.2, Q23(iv)

Given: $\tan^{-1}\left(\frac{x-2}{x-1}\right) + \tan^{-1}\left(\frac{x+2}{x+1}\right) = \frac{\pi}{4}$

Solution:

Applying the addition formula with $a = \frac{x-2}{x-1}$ and $b = \frac{x+2}{x+1}$, and combining the fractions carefully over the common denominator $(x-1)(x+1)$: expanding $(x-2)(x+1) + (x+2)(x-1)$ for the numerator of $a+b$, and $(x-1)(x+1) - (x-2)(x+2)$ for the numerator of $1-ab$, gives (after full expansion) $a+b = \frac{2x^2-4}{x^2-1}$ and $1-ab = \frac{2x^2-7}{x^2-1}$.

Dividing (the x^2-1 denominators cancel): $\frac{a+b}{1-ab} = \frac{2x^2-4}{2x^2-7}$.

Setting this equal to $\tan \frac{\pi}{4} = 1$: $2x^2-4 = 2x^2-7$ would be impossible, so re-checking the

expansion of $1-ab$ more carefully: $1-ab = 1 - \frac{(x-2)(x+2)}{(x-1)(x+1)} = \frac{(x^2-1) - (x^2-4)}{x^2-1} =$

$\frac{3}{x^2-1}$, and $a+b = \frac{(x-2)(x+1) + (x+2)(x-1)}{x^2-1} = \frac{(x^2-x-2) + (x^2+x-2)}{x^2-1} = \frac{2x^2-4}{x^2-1}$.

So $\frac{a+b}{1-ab} = \frac{2x^2-4}{3} = 1 \Rightarrow 2x^2-4=3 \Rightarrow x^2 = \frac{7}{2}$.

Taking square roots: $x = \pm\sqrt{\frac{7}{2}} = \pm\frac{\sqrt{14}}{2}$; both roots check out in the original equation by the symmetric $\pm p$ structure.

Final Answer

$$x = \pm\frac{\sqrt{14}}{2}$$

Common Mistake: Rushing the algebra when combining the two fractions $\frac{x-p}{x-q}$ and $\frac{x+p}{x+q}$ – expand each product on a separate line rather than trying to combine everything mentally, since a single sign slip anywhere changes the final quadratic.

Alternate Board-Style Example

Source: Exercise 2.2, Q23(vi)

Given: $\tan^{-1}\left(\frac{x-3}{x-4}\right) + \tan^{-1}\left(\frac{x+3}{x+4}\right) = \frac{\pi}{4}$

Solution:

Following the same method with $p=3$, $q=4$: $1-ab = \frac{(x^2-16)-(x^2-9)}{x^2-16} = \frac{-7}{x^2-16}$,

and $a+b = \frac{(x^2-7x+12)+(x^2+7x-...)}{x^2-16}$; carrying out the full expansion carefully as

in the first example and setting $\frac{a+b}{1-ab} = 1$ leads to $x^2 = \frac{17}{2}$.

Taking square roots: $x = \pm\sqrt{\frac{17}{2}} = \pm\frac{\sqrt{34}}{2}$.

Final Answer

$$x = \pm\frac{\sqrt{34}}{2}$$

Quick Recall: This whole symmetric family always reduces to $x^2 = \frac{p^2+2pq-q^2}{2}$ once you carefully expand the addition formula – both $\pm$ roots are usually valid here, unlike Type 46's asymmetric version.

Type 48

Solve $\cos(\sin^{-1}x) = k$ or $\cos(2\sin^{-1}x) = k$ for x , by converting the composite expression to an algebraic one in x using the Pythagorean identity (for the simple case) or the double-angle cosine formula (for the doubled case).

Formula Used: $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$

Formula Used: $\cos(2 \sin^{-1} x) = 1 - 2x^2$

Source: Exercise 2.2, Q23(vii)

Given: $\cos(\sin^{-1} x) = \frac{1}{9}$

Solution:

Using the identity $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$ (always non-negative, since $\sin^{-1} x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ makes its cosine non-negative), the equation becomes $\sqrt{1 - x^2} = \frac{1}{9}$.

Squaring both sides: $1 - x^2 = \frac{1}{81}$, so $x^2 = 1 - \frac{1}{81} = \frac{80}{81}$.

Taking square roots: $x = \pm \sqrt{\frac{80}{81}} = \pm \frac{4\sqrt{5}}{9}$; both signs are valid here because $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$ is an even function of x – it gives the same value for x and $-x$.

Final Answer

$$x = \pm \frac{4\sqrt{5}}{9}$$

Common Mistake: Assuming only the positive root is valid – unlike $\sin(\cos^{-1} x)$ -type expressions where a quadrant restriction sometimes forces a single sign, $\cos(\sin^{-1} x) = \sqrt{1 - x^2}$ is symmetric in x , so both roots genuinely work here.

Alternate Board-Style Example

Source: Exercise 2.2, Q23(viii)

Given: $\cos(2 \sin^{-1} x) = \frac{1}{9}$

Solution:

Using the double-angle identity $\cos(2 \sin^{-1} x) = 1 - 2x^2$, the equation becomes $1 - 2x^2 = \frac{1}{9}$.

Rearranging: $2x^2 = 1 - \frac{1}{9} = \frac{8}{9}$, so $x^2 = \frac{4}{9}$, giving $x = \pm \frac{2}{3}$.

Final Answer

$$x = \pm \frac{2}{3}$$

Quick Recall: $\cos(\sin^{-1} x)$ always converts to $\sqrt{1-x^2}$, and $\cos(2\sin^{-1} x)$ always converts to $1-2x^2$ – memorise both conversions directly rather than re-deriving them each time.

Type 49

Solve a nested composite equation like $\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$ by working from the outside in: first undo the outermost inverse function to pin down the value of the inner trig expression, then solve the remaining inner equation for x .

Source: Exercise 2.2, Q23(ix)

Given: $\cos^{-1}(\sin(\cos^{-1} x)) = \frac{\pi}{6}$

Solution:

Since $\cos^{-1}(\cdot) = \frac{\pi}{6}$ means the quantity inside must satisfy $(\cdot) = \cos \frac{\pi}{6}$ (undoing the outermost $\cos^{-1}$ first), we get $\sin(\cos^{-1} x) = \frac{\sqrt{3}}{2}$.

Now let $\cos^{-1} x = \theta$, so $\cos \theta = x$ and $\theta \in [0, \pi]$; we're told $\sin \theta = \frac{\sqrt{3}}{2}$, and since $\theta \in [0, \pi]$ where sine is always non-negative, this is consistent.

Using $\cos^2 \theta = 1 - \sin^2 \theta = 1 - \frac{3}{4} = \frac{1}{4}$, so $\cos \theta = \pm \frac{1}{2}$ – both signs are geometrically possible since θ could be in the first or second quadrant (both within $[0, \pi]$) and still have the same positive sine value.

Since $x = \cos \theta$, both values are valid solutions.

Final Answer

$$x = \pm \frac{1}{2}$$

Common Mistake: Assuming θ must be in the first quadrant only, and missing the second-quadrant solution – within $\cos^{-1}$'s range $[0, \pi]$, a given positive sine value corresponds to two possible angles, not one.

Alternate Board-Style Example

Source: Exercise 2.2, Q23(x)

Given: $\sin^{-1}(\cos(\sin^{-1} x)) = \frac{\pi}{3}$

Solution:

Undoing the outer $\sin^{-1}$: $\cos(\sin^{-1} x) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$.

Using the identity $\cos(\sin^{-1} x) = \sqrt{1-x^2}$ (non-negative, as established in Type 48):

$$\sqrt{1-x^2} = \frac{\sqrt{3}}{2}, \text{ so } 1-x^2 = \frac{3}{4}, \text{ giving } x^2 = \frac{1}{4}.$$

Final Answer

$$x = \pm \frac{1}{2}$$

Quick Recall: Peel the outermost inverse function off first by applying its matching direct trig function to both sides, then solve whatever inner equation remains – always double-check whether both signs/quadrants are genuinely possible before finalising.

Type 50

Solve $\sin^{-1}\left(\frac{a}{x}\right) + \sin^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$ for x (the variable is in the denominator), by isolating one term using the complementary identity, then squaring to eliminate the square root that appears from converting sine to cosine.

Formula Used: $\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$

Source: Exercise 2.2, Q23(xi)

Given: $\sin^{-1} \frac{8}{x} + \sin^{-1} \frac{15}{x} = \frac{\pi}{2}$

Solution:

Isolate one term: $\sin^{-1} \frac{8}{x} = \frac{\pi}{2} - \sin^{-1} \frac{15}{x}$, which sets up the complementary identity we'll need next.

Taking sine of both sides: $\frac{8}{x} = \sin\left(\frac{\pi}{2} - \sin^{-1} \frac{15}{x}\right) = \cos\left(\sin^{-1} \frac{15}{x}\right)$, using the co-function identity.

Using $\cos(\sin^{-1} u) = \sqrt{1-u^2}$ with $u = \frac{15}{x}$: $\frac{8}{x} = \sqrt{1 - \frac{225}{x^2}}$.

Squaring both sides: $\frac{64}{x^2} = 1 - \frac{225}{x^2}$, so $\frac{64}{x^2} + \frac{225}{x^2} = 1$, giving $\frac{289}{x^2} = 1$.

Therefore $x^2 = 289$, so $x = \pm 17$; since x appears in a denominator with $8/x$ and $15/x$ both needing to stay within $[-1, 1]$ and produce a positive-angle sum of $\pi/2$, the positive root $x = 17$ is the expected physical solution (matching the standard 8-15-17 Pythagorean triple).

Final Answer

$x = 17$ (taking the positive root consistent with the given positive numerators)

Common Mistake: Not recognising the 8, 15, 17 Pythagorean triple, which would have confirmed the answer instantly instead of requiring the full derivation – recognising common triples speeds up both solving and checking.

Quick Recall: This is really just $x = \sqrt{a^2 + b^2}$ in disguise – isolate one term, convert sine to cosine via the co-function identity, square to remove the square root, and simplify.

Type 51

Prove that $\sin^{-1}\left(\frac{2a}{1+a^2}\right) + \sin^{-1}\left(\frac{2b}{1+b^2}\right) = 2\tan^{-1}x$ and find x in terms of a, b , by recognising each term on the left as a disguised double-angle form of $\tan^{-1}$, reducing the whole equation to a routine addition-formula problem.

Formula Used: $\sin^{-1}\left(\frac{2t}{1+t^2}\right) = 2\tan^{-1}t, \quad |t| \leq 1$

Source: Exercise 2.2, Q23(xii)

Given: $\sin^{-1}\frac{2a}{1+a^2} + \sin^{-1}\frac{2b}{1+b^2} = 2\tan^{-1}x$, find x in terms of a, b

Solution:

Each term on the left matches the double-angle sine pattern exactly: $\sin^{-1}\left(\frac{2a}{1+a^2}\right) = 2\tan^{-1}a$ and $\sin^{-1}\left(\frac{2b}{1+b^2}\right) = 2\tan^{-1}b$, for $|a|, |b| \leq 1$.

So the left side becomes $2\tan^{-1}a + 2\tan^{-1}b = 2(\tan^{-1}a + \tan^{-1}b)$, factoring out the common 2.

Applying the addition formula inside the bracket: $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$,

so the full left side is $2\tan^{-1}\left(\frac{a+b}{1-ab}\right)$.

Comparing this to the right side $2\tan^{-1}x$, and since both are of the form " $2\tan^{-1}(\cdot)$," we can equate the arguments directly.

Final Answer

$$x = \frac{a+b}{1-ab}$$

Common Mistake: Trying to expand or square the double-angle sine expressions from first principles instead of recognising the standard $\frac{2t}{1+t^2} \rightarrow 2\tan^{-1}t$ pattern immediately – this pattern-matching step is what makes the whole problem quick.

Quick Recall: Spot $\frac{2t}{1+t^2}$ inside $\sin^{-1}$ as $2\tan^{-1}t$ in disguise, factor out the 2's, apply the ordinary addition formula inside, and read off x by comparing arguments.